



SPRINT 72

72nd BPSC Prelims Batch

Indian Geography English Notes - Part 1

Indian Geography

Part - 1

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A GENERAL INTRODUCTION TO INDIA

NOMENCLATURE & HISTORICAL NAMES

- **Bharat / Bharatvarsha:** Officially named 'Bharat'. Derived from the ancient Bharata Jana (tribe) of the Rigvedic period. Mentioned first in the Puranas (Vayu Purana names it after Bharat, son of Dushyant and Shakuntala).
- **Other Ancient Names:** 'Jambu-dvipa' and 'Aryavarta' — named 'Aryavarta' because it was the base of the Aryans.
- **Medieval Names:** Medieval Persian and Arabic historians referred to this land as 'Hind' or 'Hindustan'.
- **India:** The word 'India' used as a synonym for Bharat, is derived from the Greek word 'Indos'.

GEOGRAPHICAL LOCATION & EXTENT

India lies entirely in the **Northern** (latitudinal) and **Eastern** (longitudinal) Hemispheres. It spans both tropical and sub-tropical/warm temperate zones.

Parameter	Details
Total Area	32,87,263 sq. km (~3.28 million sq. km)
Global Share	2.4% of global land surface; holds ~17.5% of world population
World Rank (Area)	7th (Russia > Canada > USA > China > Brazil > Australia > India)
World Rank (Population)	1st (Surpassed China)
Latitudinal Extent	8°4' N to 37°6' N (Mainland) / 6°45' N to 37°6' N (Entire country)
Longitudinal Extent	68°7' E to 97°25' E
North-South Length	3,214 km (Kashmir to Kanyakumari)
East-West Width	2,933 km (Arunachal Pradesh to Gujarat)
Shape	Roughly Quadrangular

Extreme Points of India

Direction	Location	Place Name
Northernmost	Ladakh (near Siachen Glacier)	Indira Col
Southernmost (Mainland)	Kanyakumari, Tamil Nadu	Cape Comorin
Southernmost (Overall)	Great Nicobar Island (6°45' N)	Indira Point (Pygmalion Point)
Easternmost	Anjaw District {situated on LAC along the Lohit river} Arunachal Pradesh	Kibithu
Westernmost	Kutch District, Gujarat	Guhar Moti

MAJOR GEOGRAPHIC LINES**Indian Standard Time (IST)**

- **Standard Meridian:** 82°30' E Longitude (Passes through Mirzapur/Naini, Uttar Pradesh).
- **Time Difference:** It takes 4 minutes to cross 1 degree of longitude, which is why Indian time is ahead of GMT by 5 hours and 30 minutes. The East-West time gap in India is approximately 2 hours (the sun rises 2 hours earlier in Arunachal Pradesh than in Gujarat).
- **Passes through 5 States:** Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha, Andhra Pradesh.

Tropic of Cancer (23°30' N)

- Divides India into two nearly equal halves.
- **Passes through 8 States (West to East):** Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram.
- The **Mahi River** in India is unique because it is the only river that crosses the **Tropic of Cancer** twice. It crosses the line first in Madhya Pradesh and then again in Gujarat.
- **Intersection:** IST and the Tropic of Cancer intersect in Koriya District (Baghelkhand Region), Chhattisgarh.

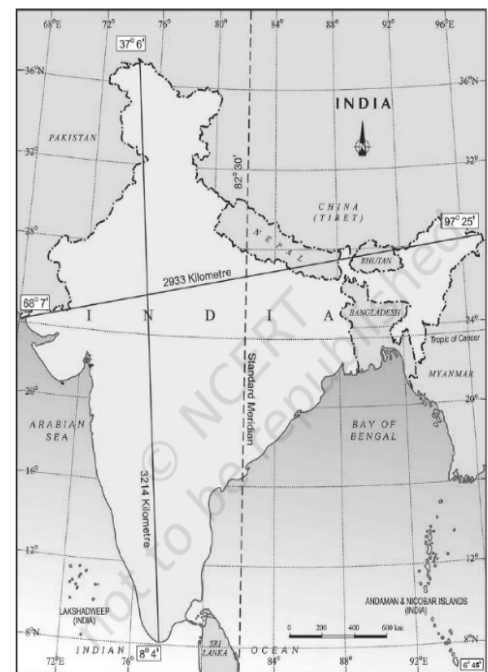


Figure 1.3 : India : Extent and Standard Meridian

COASTLINES, LAND BORDERS & NEIGHBORS**International Borders**

India shares a land boundary of approximately **15,106.7 km** (MHA) with 7 countries. 16 States and 2 UTs share international borders.

Country	Length (km)	Bordering Indian States / UTs
Bangladesh	4,096.7	West Bengal, Assam, Meghalaya, Tripura, Mizoram
China	3,488.0	Ladakh, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh
Pakistan	3,323.0	Gujarat, Rajasthan, Punjab, J&K, Ladakh
Nepal	1,751.0	Uttarakhand, Uttar Pradesh, Bihar, West Bengal, Sikkim

Myanmar	1,643.0	Arunachal Pradesh, Nagaland, Manipur, Mizoram
Bhutan	699.0	Sikkim, West Bengal, Assam, Arunachal Pradesh
Afghanistan	106.0	Ladakh (Wakhan Corridor / PoK)

Borders listed in descending order of length.

Important Border Lines & Water Channels

- **Radcliffe Line:** India–Pakistan & India–Bangladesh (Superimposed boundary).
- **McMahon Line:** India–China (drawn by Sir Henry McMahon during the Shimla Convention on July 3, 1914.).
- **Durand Line:** India–Afghanistan (drawn by Mortimer Durand, 1893).
- **Palk Strait & Gulf of Mannar:** Separates India (Tamil Nadu) and Sri Lanka.
- **8° Channel:** Maldives — Minicoy (India).
- **9° Channel:** Minicoy — Lakshadweep.
- **10° Channel:** Andaman — Nicobar.
- **Great Channel:** Indira Point — Indonesia.

Coastline Facts

- **Total Coastline:** 11098.81 km (9 States and 4 UTs have coastlines).
- **State Coastline Rank (Descending):** Gujarat > Tamil Nadu > Andhra Pradesh > Maharashtra > West Bengal.
- **Territorial Sea Limit:** 12 Nautical Miles (~21.9 km). (68th BPSC PYQ)

ADMINISTRATIVE DIVISIONS & KEY UT FACTS

India has **28 States and 8 Union Territories**.

Notable UT Integrations Post-Independence

- **Portuguese Colonies (1961):** Goa, Daman & Diu, Dadra & Nagar Haveli.
- **French Colonies (1954):** Puducherry, Karaikal, Mahe, Yanam, Chandannagar.

Puducherry is uniquely spread across three states: Yanam (Andhra Pradesh), Karaikal & Main Puducherry (Tamil Nadu), Mahe (Kerala).

Zonal Councils

Established under the States Reorganisation Act, 1956 (except the North-Eastern Council, created under the NEC Act, 1971). Zones: Northern, Eastern, Western, Central, Southern, and North-Eastern.

National Capital Region (NCR)

Includes Delhi and surrounding districts: Haryana (14 districts), Uttar Pradesh (8 districts), and Rajasthan (2 districts — Alwar, Bharatpur).

MISCELLANEOUS

- **Sonbhadra (Uttar Pradesh):** The only district in India that touches four states: Bihar, Jharkhand, Chhattisgarh, and Madhya Pradesh.

Tri-Junction States

- **Sikkim:** Nepal, China, Bhutan.
- **West Bengal:** Nepal, Bhutan, Bangladesh.
- **Arunachal Pradesh:** Bhutan, China, Myanmar.
- **Ladakh:** China, Pakistan, Afghanistan.

Landlocked Regions

19 States and 4 UTs are completely landlocked. Five states — Haryana, Chhattisgarh, Telangana, Madhya Pradesh, Jharkhand — have zero international borders or coastlines.

Geography Superlatives

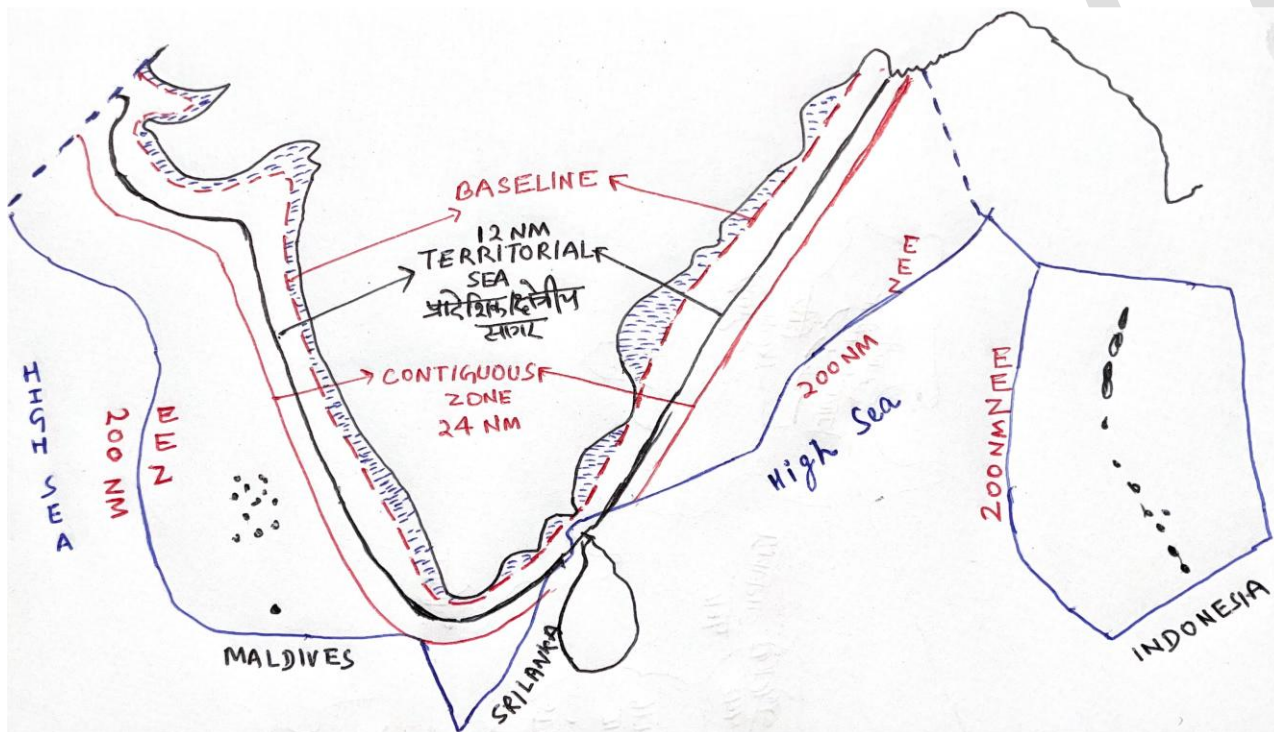
Category	Details	Note
Largest State (Area)	Rajasthan	Smallest: Goa
Largest State (Population)	Uttar Pradesh	Smallest: Sikkim
Largest District	Kutch, Gujarat	—
Lowest Point	Kuttanad, Kerala (2.2 m below sea level)	—
Highest Waterfall	Kunchikal Falls (455 m), Varahi River, Karnataka, it is ranked as the 116 th highest waterfall in the world	—
Most State Borders	Uttar Pradesh — 8 states + 1 UT (Delhi)	—

TERRITORIAL SEA & UNCLOS

UNCLOS — Overview

UNCLOS (United Nations Convention on the Law of the Sea) is an international agreement that determines the rights and responsibilities of coastal countries on the world's seas and oceans, and establishes rules for the use of marine resources. (UPSC PYQ)

Adopted by the UN	10 December 1982
Came into force	16 November 1994
Adopted by India	1995



Base Line

The **Base Line** (Low-Water Line) is the line joining the inland part to the sea. Coastal countries have the right to determine their own baseline.

Maritime Zones

Zone	Distance from Baseline	Key Rights / Jurisdiction
Internal Waters	0 (landward of baseline)	Full sovereignty; no innocent passage
Territorial Sea	0 – 12 nautical miles	Full sovereignty; innocent passage allowed
Contiguous Zone	12 – 24 nautical miles	Enforce fiscal, immigration, sanitary & customs laws (no airspace rights)

EEZ	0 – 200 nautical miles	Sovereign rights to explore, exploit, conserve & manage resources; energy; artificial islands; marine research
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1 Nautical Mile = 1.852 km / 1.15 miles

Internal Waters

The sea water between the land and the baseline is called **Internal Waters**. The coastal state exercises complete sovereignty here — foreign vessels have no right of innocent passage.

Territorial Sea (0 – 12 Nautical Miles)

The area extending up to **12 nautical miles from the baseline** is called the Territorial Sea. (68TH BPSC PYQ)

Rights of the Coastal State

Complete sovereignty over:

- Air space above the sea
- The entire sea area
- The sea floor (seabed)
- The subsoil beneath the seabed
- Exclusive rights over all resources within this zone

Innocent Passage

Foreign ships are permitted **innocent passage** provided the passage is not prejudicial to the peace, order, or security of India.

Rules for Foreign Ships in Indian Territorial Waters

- All foreign ships shall have innocent passage as long as it is not prejudicial to the peace, order or security of India.
- Foreign warships (including submarines and underwater vehicles) may enter only after giving **prior notice to the Central Government**.
- Submarines must **sail on surface and display their flag** while passing through the territorial sea.
- The Central Government may suspend entry of foreign ships into specified areas by notification in the Official Gazette.

Contiguous Zone (12 – 24 Nautical Miles)

The area up to **24 nautical miles from the baseline** is called the Contiguous Zone.

Rights in the Contiguous Zone

Coastal countries may prevent violations of:

- Fiscal laws
- Immigration laws
- Sanitary laws
- Customs laws

Note: These rights do NOT extend to airspace or outer space above the Contiguous Zone.

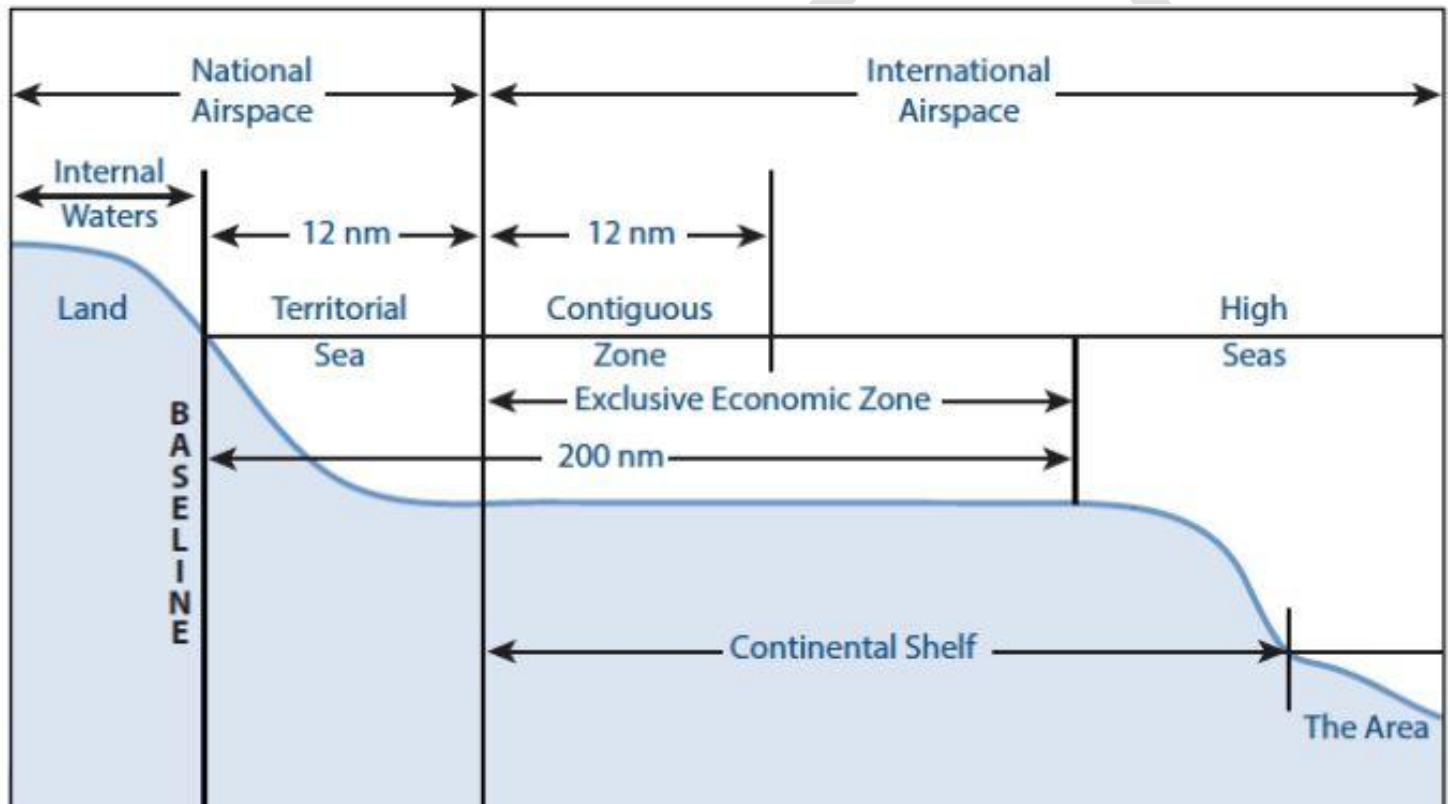
Exclusive Economic Zone — EEZ (0 – 200 Nautical Miles)

The EEZ extends up to **200 nautical miles from the baseline**. The coastal country has sovereign rights over a wide range of activities within this zone.

Rights in the EEZ

- To **explore, exploit, conserve and manage** all natural resources (living and non-living) in the water column, seabed and subsoil
- To produce energy from natural resources including **winds, tides and waves**
- To build, maintain and use **artificial islands** and offshore terminals
- To conduct **marine scientific research**
- To conserve marine resources and **prevent marine pollution**

Key Distinction: In the EEZ, the coastal state has sovereign rights (not full sovereignty). Other states retain freedoms of navigation, overflight, and laying of submarine cables/pipelines.



PHYSIOGRAPHIC REGIONS OF INDIA

India is a large country situated on the northern tip of the Indian Ocean which is full of landform diversity. Due to its vastness, it is called a subcontinent.

A large area in the north has rugged topography which includes the Himalayan mountain ranges, numerous peaks, beautiful valleys and canyons.

South India is a stable but fragmented plateau where there is an abundance of eroded rock fragments and escarpments. (a very steep piece of ground that separates an area of high ground from an area of lower ground) Between these two lies the vast plain of North India.

NORTH AND NORTH-EASTERN MOUNTAIN RANGES

The northern and north-eastern mountain ranges are classified into two parts —

Western Himalayas and Eastern Himalayas. The Western Himalayas are further divided into Trans Himalayas and Himalayan Ranges. The Eastern Himalayas include the north-eastern hills which extend from north to south.

Trans Himalayas

Situated to the north of the Himalayan ranges, the Trans Himalayas include four mountain ranges:

- Karakoram, Ladakh, Zaskar, Kailash

Karakoram is the highest mountain range of Trans Himalaya. It contains India's highest peak and the world's second highest peak **K2 / Godwin Austen (8,611 m)**, and the world's highest military base **Siachen (7,560 m)**. Strategically it is very important for India.

Major glaciers: **Siachen, Sasaini, Hispar, Biafo, Baltoro**. Also called the backbone of Upper Asia and Krishnagiri Mountain. Average width: 110–130 km.

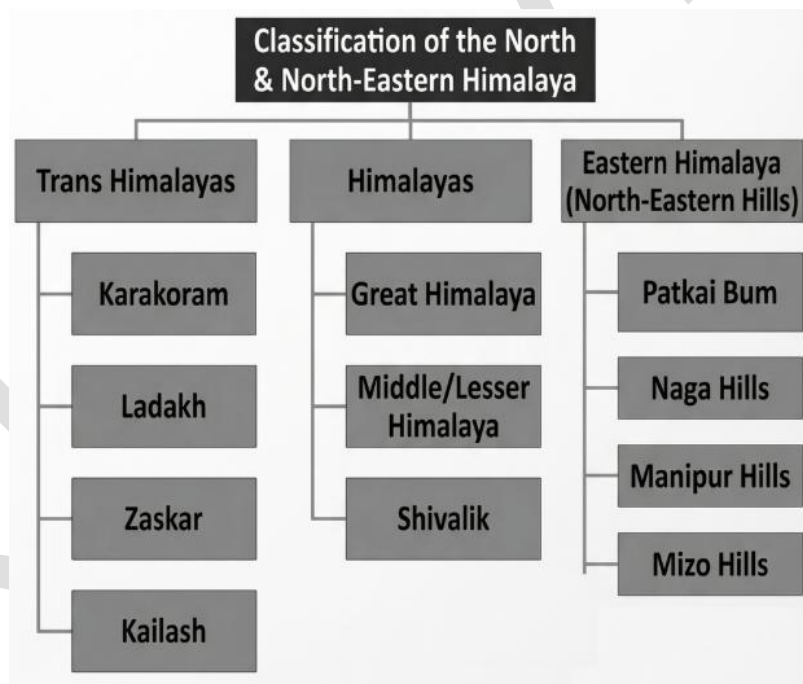
Ladakh Range lies between the Shyok and Indus rivers (SE of Karakoram). Average height: 3,500–4,000 m. Made of granite and gneiss rocks.

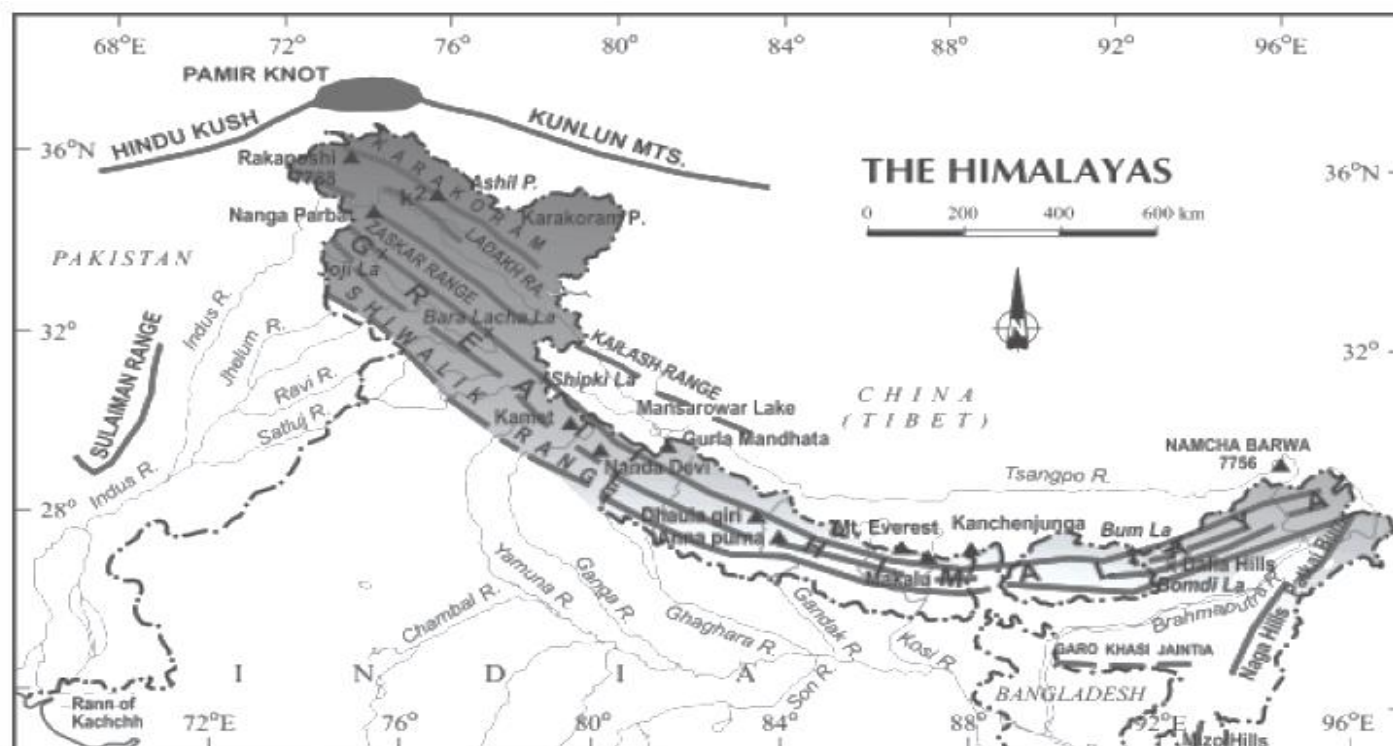
Zaskar Range lies south of Ladakh and north of the Great Himalayas. Average height: 6,000 m. The Indus flows between the Ladakh and Zaskar ranges.

Kailash Range is the eastern extension of the Ladakh range into the Tibetan region. Highest peak: Mount Kailash (6,714 m). CHRONOLOGY OF MOUNTAIN RANGES IS IMPORTANT

Himalayan Ranges

The Himalayan Mountains (FOLD MOUNTAINS) are a complex mountain system containing rocks from the Pre-Cambrian to the Eocene era (mostly sedimentary and metamorphic). Their origin is **explained by the Plate Tectonics Theory** — the Himalayas originated from the convergence of the Eurasian plate and the Indian plate, folding the sediments of the Tethys Sea. This convergence continues at a rate of increase in height of 5 mm/year.





Extent: From the Indus River Gorge (Nanga Parvat) in the west to the Brahmaputra River Gorge (Namcha Barwa) in the east — a length of 2,500 km, width 160–400 km.

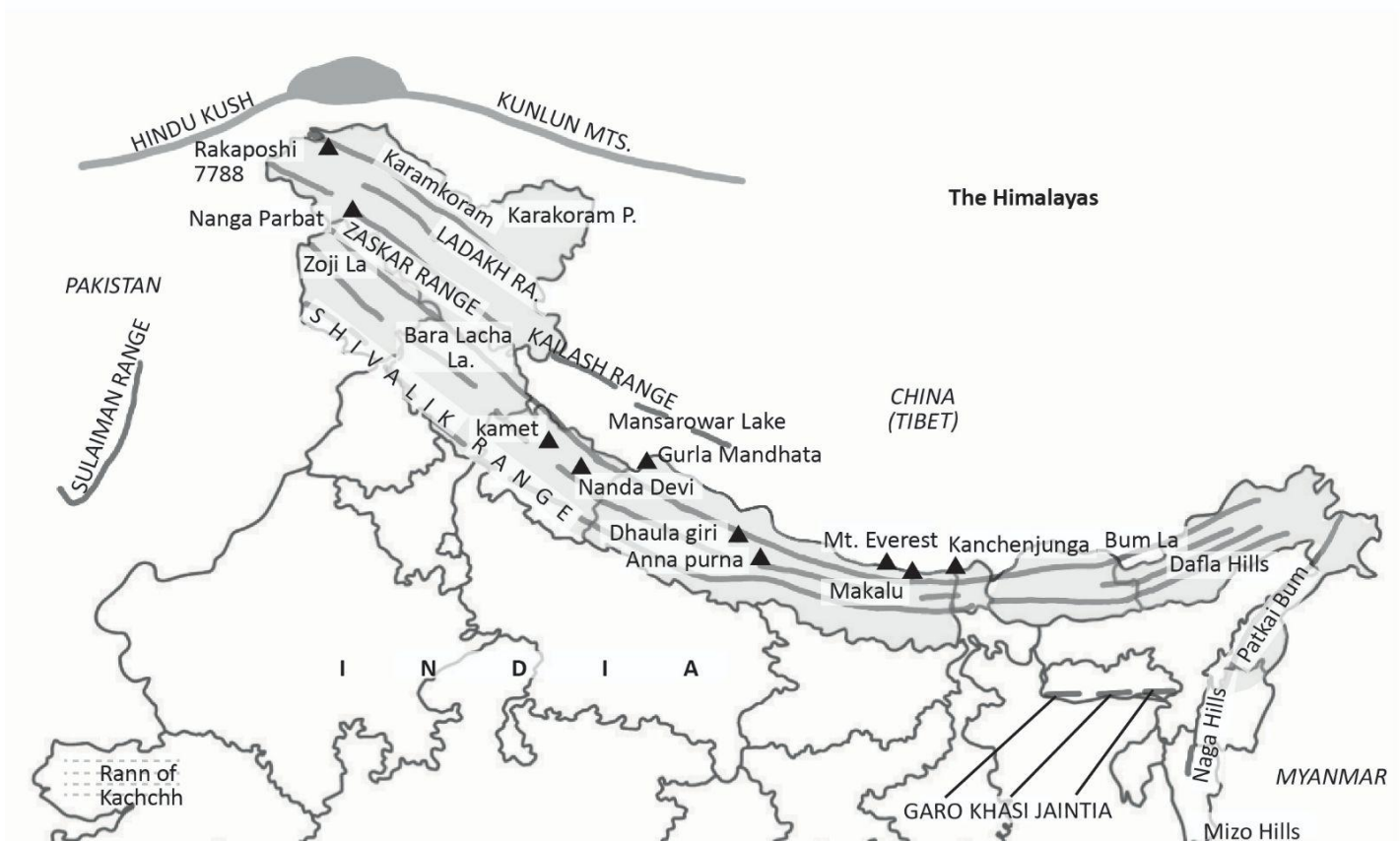
Bow-shaped with convex bend southward. Average height 6,000 m. Out of 94 Asian peaks above 7,300 m, 92 are in the Himalayas and Karakoram.

The Himalayan range consists of three parallel sub-ranges:

Sub-Range	Dimensions	Key Features / Regional Names
Great Himalaya (Himadri / Central Axial Range / Inner Himalayas)	~2,500 km long; 120–190 km wide; Avg. height 6,100 m	Mt. Everest (8,848.86 m) (The world's highest peak), Kanchenjunga (8,598 m) (World's third highest and India's second highest peaks are located in this mountain range)
Middle / Lesser Himalaya (Himachal)	Height 3,700–4,500 m; Avg. width 50 km	Pir Panjal (J&K), Dhauladhar (HP), Mussoorie/Nagtibba (Uttarakhand), Mahabharata (Nepal); hill stations: Shimla, Manali, Mussoorie, Darjeeling
Shivalik (Outer Himalayas)	Height 600–1,500 m; Width 10–50 km	Jammu Hills (Jammu), Dafia/Miri/Abor/Mishmi Hills (Arunachal); Doon valleys (west), Duar/Dwar (east)

Small grasslands on Middle/Lesser Himalaya slopes are called Marg in Kashmir (e.g. Sonamarg, Gulmarg) and Bugyal/Payar in Uttarakhand.

Shivalik deposits of coarse pebbles/stones are called Doon (Dehradun, Kotlidun, Patlidun) in the west and Duar/Dwar in the east. Sankos River divides Duar into Western and Eastern. Duar soil is fertile — home to tea gardens.



Geological Faults between Himalayan Ranges

Faults caused by the convergence process make the Himalayan mountain range one of the most earthquake-prone regions in the world:

Between	Fault
Tibetan Plateau — Great Himalayas	ITSZ (Indus-Tsangpo Suture Zone)
Great — Middle Himalayas	MCT (Main Central Thrust)
Middle/Lesser Himalayas — Shivalik	MBT (Main Boundary Thrust)
Shivalik — Shivalik Foothills	HFF (Himalayan Frontal Fault)

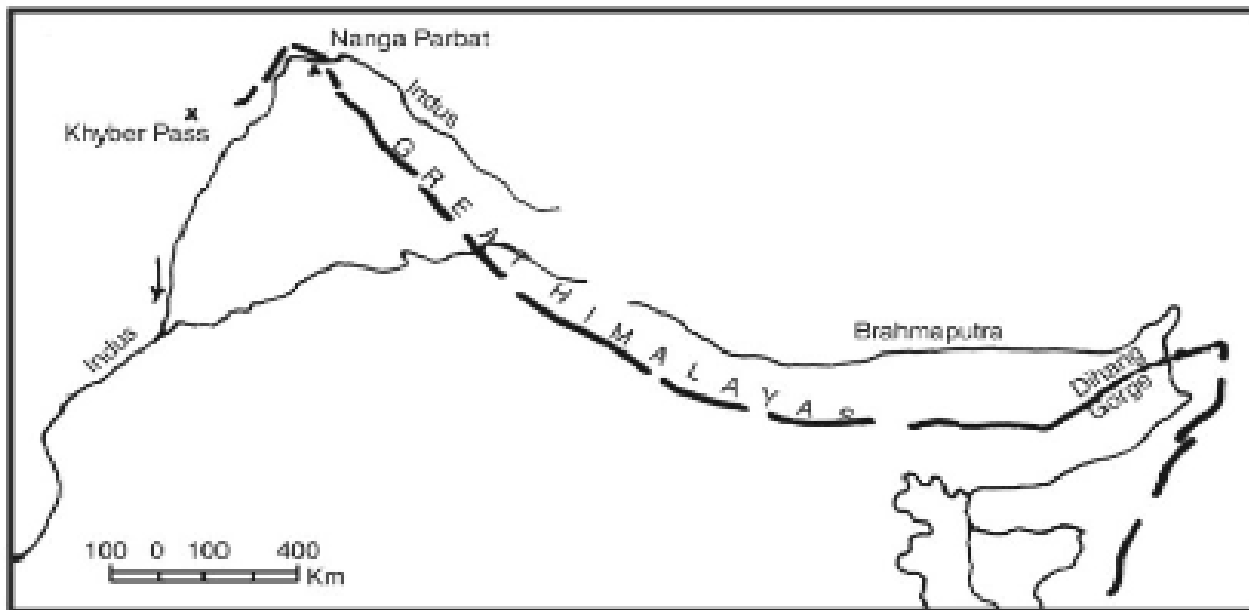
Intermountain Valleys

- **Between Great Himalaya & Lesser Himalaya:** Kashmir Valley; Lahaul Spiti Valley (HP); Valley of Flowers, Chamoli (Uttarakhand); Kathmandu Valley (Nepal).
- **Between Lesser Himalaya & Shivalik:** Kullu Valley (HP); Doon Valley (Uttarakhand).

The moraine deposit structure in the Kashmir valley is called Karewa, famous for saffron cultivation.

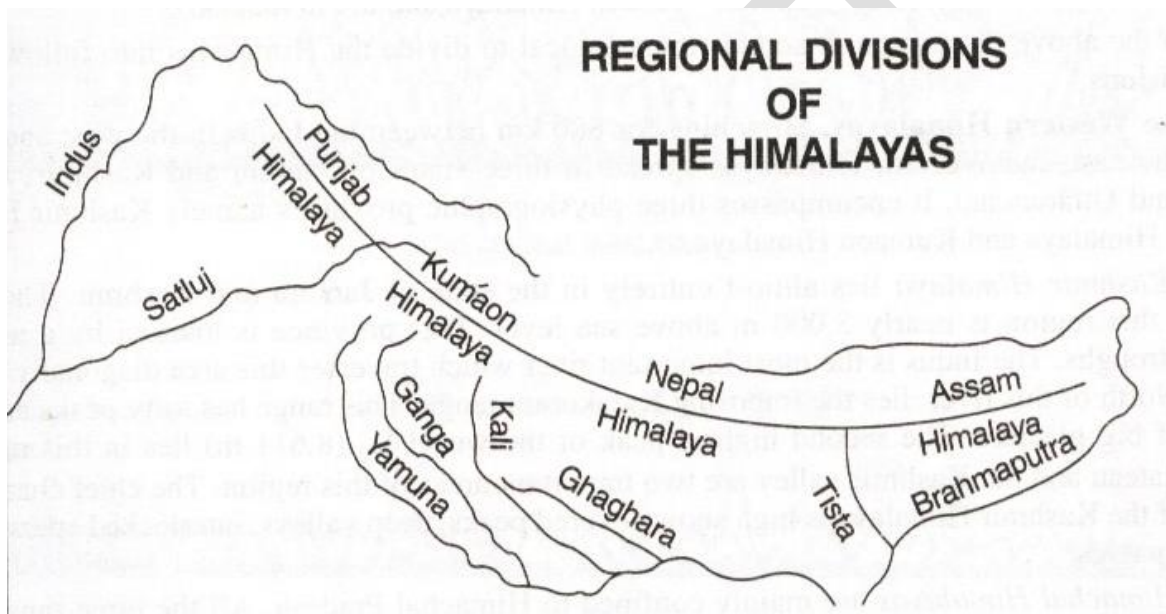
Syntaxial Bends of the Himalaya

The general extension of the Himalayas is west to east, but sharp southward bends occur at both ends. These are called syntaxial bends — near Nanga Parbat in the west and east of the Brahmaputra in Arunachal Pradesh. Both formed due to tectonic strike.



Regional Classification of the Himalaya

Concept given by Sydney Gerald Burrard. Classification is based on rivers.



Division	Extent	States / Region	Key Features
Punjab Himalaya (Kashmir / Himachal Himalaya)	560 km: Indus Gorge → Sutlej Gorge	J&K, Ladakh, Himachal Pradesh	Karakoram, Ladakh, Jaskar, Pir Panjal, Dhauladhar
Kumaon Himalaya	320 km: Sutlej Gorge → Kali Gorge	Uttarakhand	Gangotri, Yamunotri, Badrinath, Kedarnath, Nanda Devi
Nepal Himalaya	800 km: Kali Gorge → Teesta Gorge	Nepal	Highest part; Annapurna, Dhaulagiri, Makalu, Everest
Assam Himalaya	720 km: Teesta Gorge → Brahmaputra Gorge	Sikkim, Bhutan, Arunachal Pradesh	—

Eastern Himalayas (North-Eastern Hills / Purvanchal)

The Purvanchal ranges are located in the northeastern part of India, south and south-east of the Brahmaputra valley — a result of the marginal effect of the north-east movement of the Indian plate. Height gradually decreases from north to south.



Range	Location	Key Facts
Patkai Bum	Eastern Arunachal Pradesh	Biodiversity & dense forests; Namdapha National Park
Naga Hills	Nagaland	Watershed = India-Myanmar border; Highest peak: Saramati (3,826 m)
Manipur Hill	Manipur	Sloping hill; Loktak Lake (Only place in India where Phundis are found); Keibul Lamjao (world's only floating national park); Sangai deer (DANCING DEER)
Mizo Hills (Lushai Hills)	Mizoram	Highest peak: Blue Mountain (2,157 m)
Tripuri Hills	Tripura	Long ranges in the southern part of the northeastern range

Snow Line

The height above which snow remains frozen throughout the year.

Factors: latitudinal position, local topography, rainfall, slope.

Eastern Himalayas receive more rainfall (closer to Bay of Bengal) → snow line is lower.

Western Himalayas receive less rainfall → snow line is higher.

Rule: Higher humidity = Lower snow line.

Himalayan Region	Snow Line Height
North-Eastern Himalayas (Arunachal / Sikkim)	4,400 m
Nepal Himalaya	4,500 m
Kumaon Himalaya / Uttarakhand	5,200 m
Kashmir Himalaya	5,800 m
Western Himalayas (average)	5,800 m
Eastern Himalayas (average)	4,300 m

Glaciers

A large mass of moving/flowing ice. Material eroded by glaciers = moraine. Three-quarters of all fresh water on Earth is in glaciers.

- **Continental Glaciers:** Thick ice sheets covering large polar land areas (Greenland, Antarctica, Arctic). Spread outward from the centre.
- **Mountain / Valley Glaciers:** Snow mass moving downward in existing valleys. World's largest outside polar regions: Fedchenko Glacier, Tajikistan (77 km). India's largest: Siachen Glacier, Karakoram (75 km).

Major Indian Glaciers (descending order): Siachen > Sasaini > Hispar > Biafo > Baltoro

Other glaciers: Gangotri (Ganga), Yamunotri (Yamuna), Milam (Sharda), Pindari, Chorabari (Mandakini), Zemu (Sikkim — largest in NE India)

BCW BITS: Siachen Glacier is situated to the north of Nubra valley (UPSC PYQ)

Importance of the Himalayas for India

Aspect	Details
1. Climatic Influence	Blocks SW monsoon → heavy interior rainfall. Shields India from cold Central Asian winds. Controls the Jet Stream → snowfall in higher Himalayas and Rabi crop rainfall in NW India.
2. Defence	Natural barrier — no invasion from the north in Indian history. Acts as a natural defence line against China, Tibet, Nepal, and Bhutan.
3. Perennial Rivers	Glaciers, springs and lakes of Himalayas are the source of most north Indian perennial rivers, sustaining millions of livelihoods.
4. Fertile Soils	Rivers carry vast alluvial sediments forming the fertile North Indian Plains — the country's most productive agricultural region.
5. Hydroelectricity	Numerous HEP projects: Bhakra-Nangal, Nathpa Jhakri, Uri, Dulhasti, Baglihar, Salal, Tehri, Koteswar.
6. Forest Wealth	Tropical, temperate and coniferous forests yield timber, gum, lac, herbs, medicinal plants, and industrial raw materials.
7. Orchards	Apple, apricot, almond, walnut, pear, peach, cherry orchards on slopes. Saffron cultivated in Karewa land of Kashmir.
8. Mineral Resources	Limestone, iron, copper, lead, zinc, gold, coal, mica in HP, Uttarakhand, Sikkim, Arunachal. Lithium found in Reasi district (J&K). Coal near Kalakot (J&K).
9. Tourism	Srinagar, Pahalgam, Gulmarg, Shimla, Manali, Mussoorie, Darjeeling, Dehradun and many more attract millions of tourists yearly.

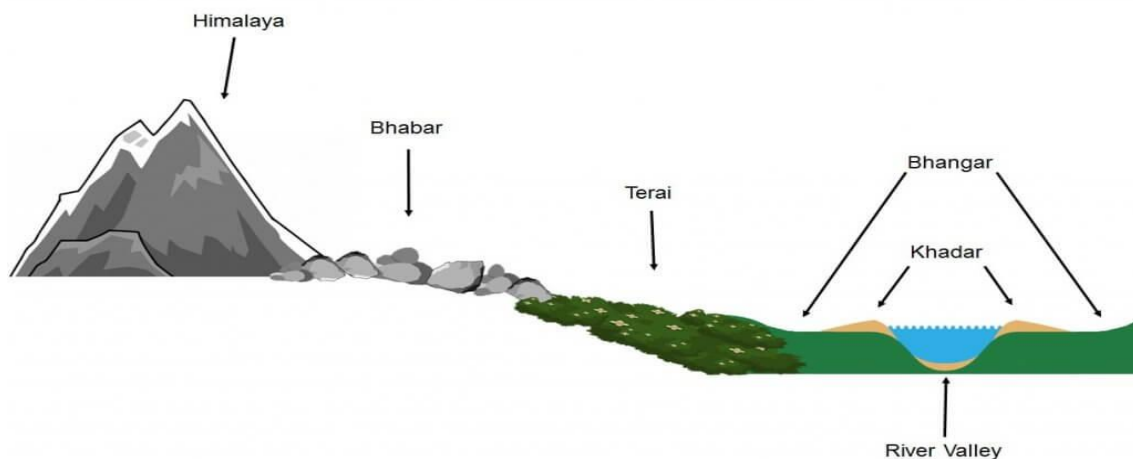
10. Pilgrimage Sites

Amarnath, Vaishno Devi, Kedarnath, Badrinath, Gangotri, Yamunotri, Hemkund Sahib, Tawang Monastery, and others.

NORTH INDIAN PLAIN

Formation: Alluvial deposit formed by the Indus, Ganges, Brahmaputra and their Himalayan tributaries in the trench created by the subduction of the Indo-Australian Plate under the Eurasian Plate. Situated south of the Himalayas and north of the peninsular plateau.

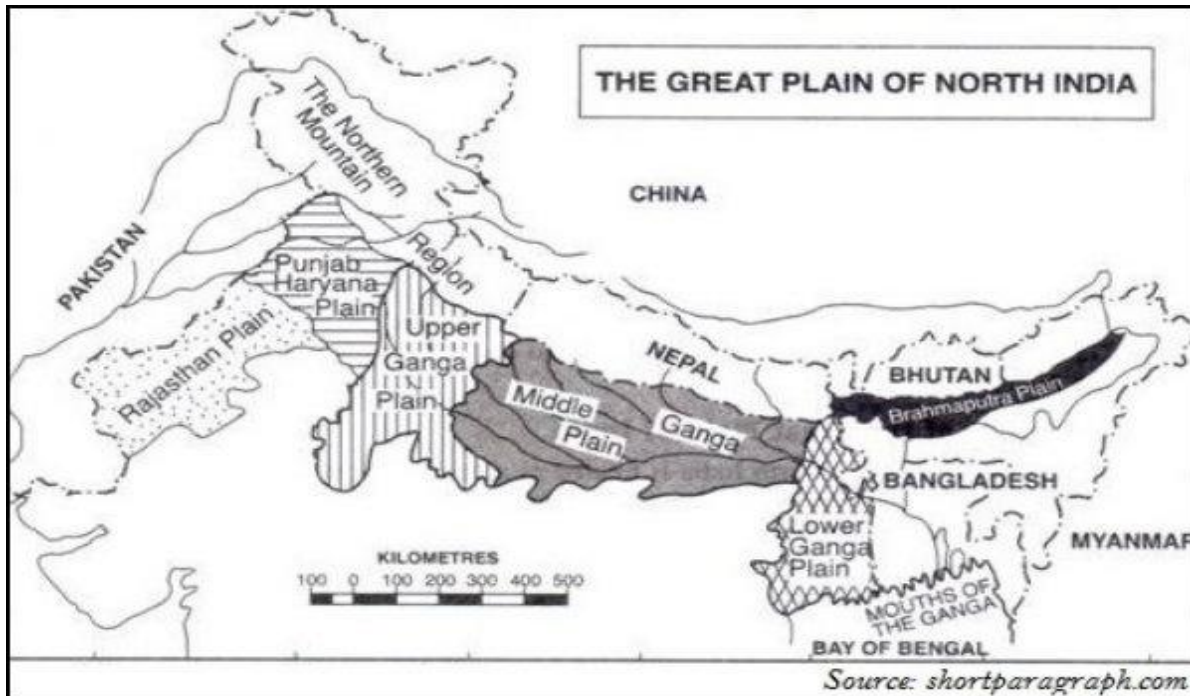
Parameter	Details
Area	~7.5 lakh sq. km
Length (E–W)	~3,200 km
Width (N–S)	150–400 km (narrowest in Assam: 90–100 km; widest in West Bengal: up to 400 km)
Avg. Height	150–200 m above sea level
Max. Alluvial Depth	1,000–2,000 m

North–South Zones of the Plain

Zone	Location	Key Features
Bhabar	8–10 km strip parallel to Shivalik foothills	Rivers deposit heavy sediments (soil, sand, pebbles) → permeable rocks → rivers disappear here. Unsuitable for farming; good for deep-rooted trees.
Terai	10–20 km south of Bhabar	Rivers reappear; no definite channels → swampy. High water table. Natural vegetation, wildlife. Paddy and sugarcane cultivated after deforestation.
Alluvial Plain (Bhangar)	Higher ground; flood water does not reach	Older alluvium. Sutlej Plain & Upper Gangetic Plain. Fertile but excessive irrigation in dry areas → Kallar / Usar / Reh / Thur (saline/alkaline soil) in W. UP and Haryana.
Alluvial Plain (Khadar)	Low-lying; flooded almost every year	Newer alluvium; fine clay soil. Very fertile; rice, wheat, maize, jute, sugarcane. Found in eastern UP, N. Bihar, Hooghly coast (WB), Assam. Excellent groundwater receptor.

Deltaic plain (W. Bengal) — higher parts = char; lower moist submerged land = beel. Sundarban Delta = world's largest delta (Ganga + Brahmaputra), also world's largest single-patch mangrove forest.

Regional Divisions of the North Indian Plain



Northern Rajasthan Plain

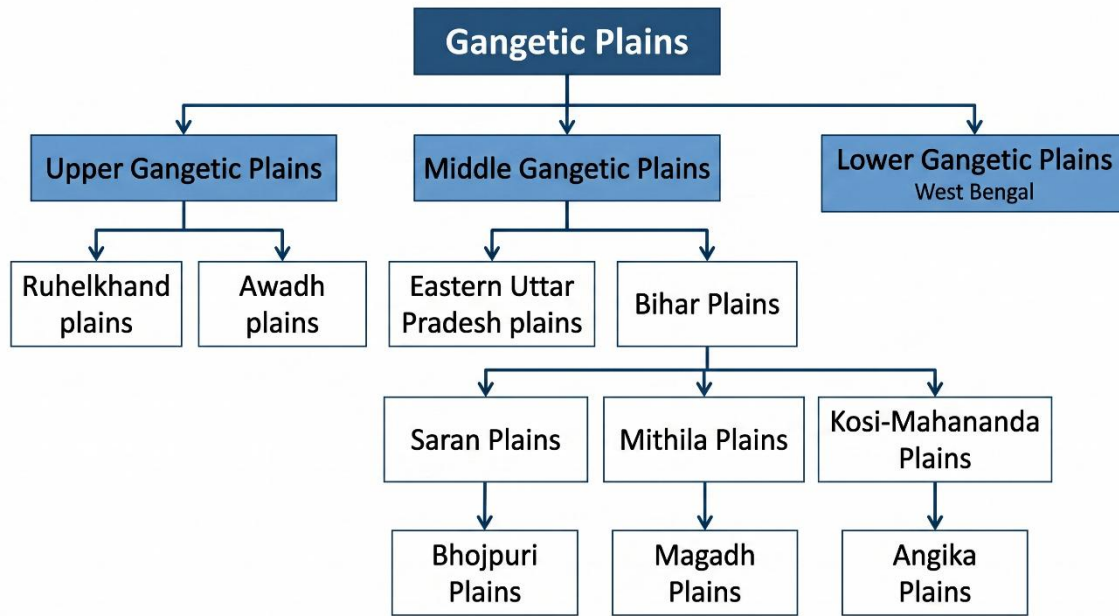
Extends from west of the Aravalli range to the India–Pakistan border. Length (NE–SW): 640 km; Area: 1.75 lakh sq. km. Divided into: Desert (SW, Thar sand dunes) and Rajasthan Bangar (NE, former Ghaggar river channel, fertile).

Punjab-Haryana Plain

Area ~1.75 lakh sq. km, height 200–300 m. Formed by the Indus and tributaries. Made up of doabs:

Doab Name	Between Rivers
Sindh Sagar Doab	Indus — Jhelum
Chaj Doab	Jhelum — Chenab
Rachna Doab	Chenab — Ravi
Bari Doab	Ravi — Beas (Amritsar, Lahore)
Bist Doab	Beas — Sutlej (Jalandhar, Hoshiarpur)

Gangetic Plain



Extends across UP, Bihar and West Bengal. Slopes NW to SE. Formed by Ganga and tributaries (Yamuna, Ramganga, Sharda, Gomti, Ghaghra) plus peninsular rivers (Chambal, Betwa, Ken, Son).

Sub-Division	Extent / Boundaries	Key Features
Upper Gangetic Plain	Two-thirds of UP; bounded by Shivalik (N), peninsular plateau (S), Yamuna (W), 100-m contour (E)	Ganga-Yamuna Doab, Rohilkhand Plain, Awadh Plain; rice, wheat, sugarcane, maize, pulses; Green Revolution success.
Middle Gangetic Plain	~1.45 lakh sq. km; E. UP and Bihar; up to 150 m high	Ghaghra, Gandak, Kosi, Son flow here. Kosi ('Sorrow of Bihar') infamous for floods (shifted 96 km W since 1936). Sub-regions: Ganga-Ghaghra Doab, Magadha Plain, etc. Dense population; intensive farming.
Lower Gangetic Plain	Mainly West Bengal; narrows to 16 km near Rajmahal Hills	Slope 2 cm/km in delta → Ganga splits into streams → Sundarbans Delta (world's largest delta, mangrove forest). Fertile; rice and jute.

Assam / Brahmaputra Plain

Almost entirely in Assam (also called Assam Plain). Surrounded by Himalayas (N), Patkai & Naga ranges (E), Garo-Khasi-Jaintia Hills (S), Lower Gangetic Plain / India-Bangladesh border (W). Length: ~820 km (Dhubri → Sadiya). Slopes NE to SW. Features: oxbow lakes, sand dunes, riverine islands. Home to Kaziranga National Park (one-horned rhinoceros) and Manas Biosphere Reserve.

Significance of the Great Plains of India

Aspect	Details
Fertile Soils	Supports all tropical and temperate crops; both cereals and cash crops.
Perennial Rivers	Year-round water supply; canal irrigation provides sustainability.
Groundwater	Abundant groundwater; tube-well irrigation widely used.
Navigation	Gentle slope → rivers easily navigable by boat.
Infrastructure	Road, highway and railway networks easily developed on flat terrain.

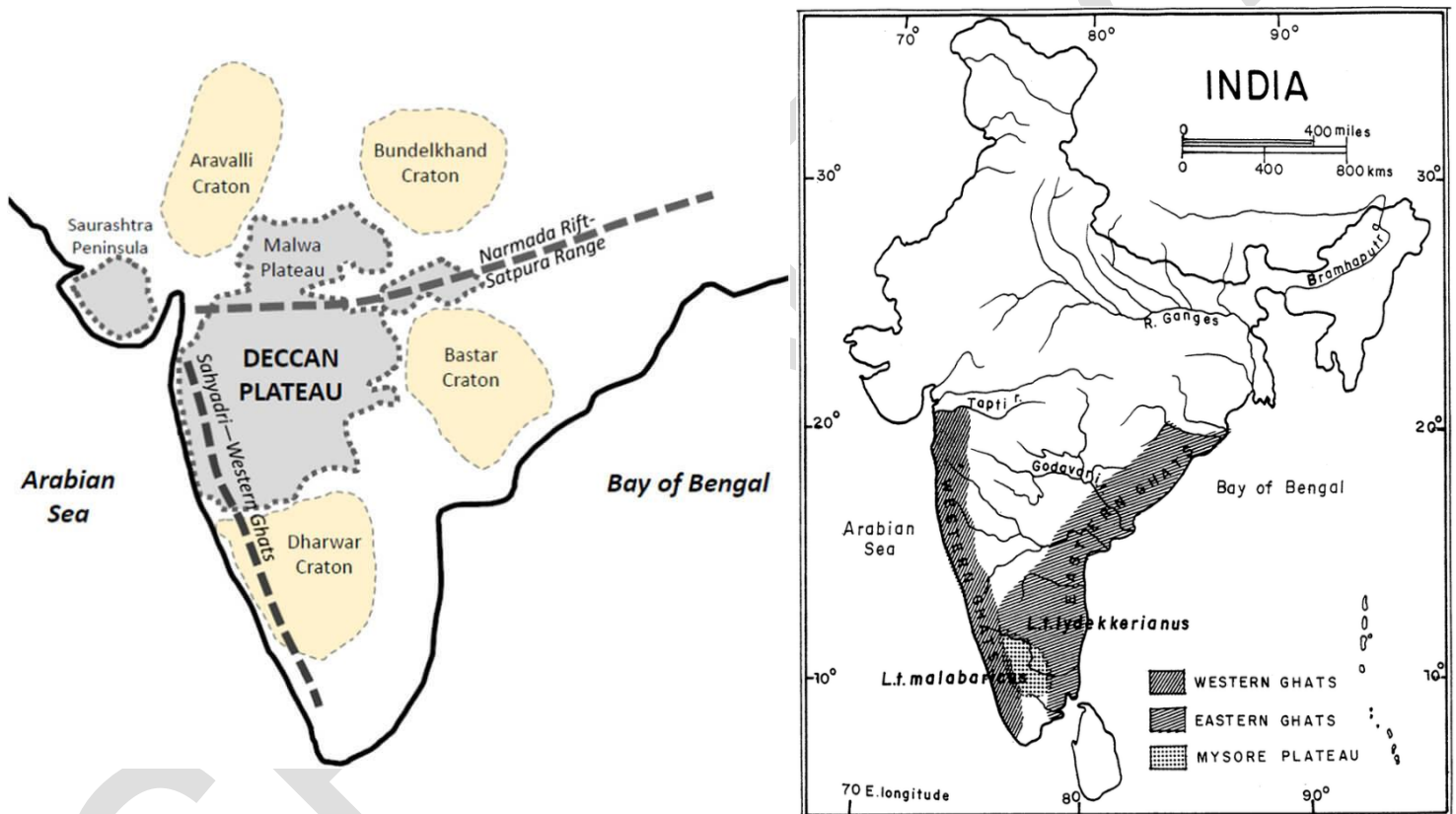
Petroleum & Gas	Reserves found in sedimentary rocks of North India.
Population	~23% of India's total area but ~40% of its population; most densely farmed region.
Cultural Heritage	Major pilgrimage sites of Hinduism, Buddhism, Jainism, Sikhism and Islam: Amritsar, Haridwar, Prayagraj, Varanasi, Kushinagar, Patna, Bodh Gaya.

PENINSULAR PLATEAU

Triangular, fragmented landmass rising 150 m above river plains. Average height: 600–900 m. Boundaries: Aravalli extension (Delhi Ridge, NW), Rajmahal Hills (E), Gir Hills (W), Cardamom Hills (S). Also includes Meghalaya Plateau and Karbi-Anglong Plateau (NE). One of the oldest and most stable landforms of India.

General slope: West to east (confirmed by eastward-flowing rivers like Godavari).

Plateaus: Malwa, Bundelkhand, Baghelkhand, Chotanagpur, Chhattisgarh, Dandakaranya (Chhattisgarh BPSC PYQ), Telangana, Deccan, Rayalaseema, Karnataka.



(i) Deccan Plateau

One of the oldest landforms; formed by cooling and solidification of basaltic lava. Area: ~5 lakh sq. km. **Bounded by Western Ghats (W), Eastern Ghats (E), Satpura (N).** Mainly Maharashtra + parts of Karnataka, Telangana, AP, Tamil Nadu, MP, Chhattisgarh, Jharkhand. Main hills: Satmala, Ajanta, Harishchandra, Balaghat.

Western Ghats

Length: 1,600 km (N–S); **Width:** 50–80 km; **Avg. Height:** 1,500 m (increases N to S). Area: ~1,60,000 sq. km.

Regional names: Sahyadri (Maharashtra), Nilgiri (Karnataka & Tamil Nadu), Anamalai & Cardamom Hills (Kerala).

Feature	Name / Value	Location / Note
Highest peak of peninsular plateau	Anaimudi (2,695 m)	Anamalai Hills, Western Ghats
Second highest peak	Doddabetta (2,637 m)	Nilgiri Hills
Famous hill station	Udhagamandalam / Ooty ('Queen of Hill Stations')	Nilgiri Hills (TN–Kerala–Karnataka border)
Special flower (blooms every 12 yrs)	Neelakurinji	Nilgiri Hills
Eastern extension of Anamalai Hills	Palani Hills	Tamil Nadu; Kodaikanal hill station
Major passes	Thalghat, Bhorphat, Palghat, Senkota	Roads & railways access

Western Ghats: UNESCO World Heritage Site; one of 36 global biodiversity hotspots (and one of the 8 most prominent).

Eastern Ghats

Area: ~75,000 sq. km; Avg. height: 600 m. Not continuous — eroded by Mahanadi, Godavari, Krishna and Kaveri. Main ranges: Mahendragiri, Udayagiri, Nallamalla, Velakonda, Seshachalam, Palakonda, Javadi, Shivarai, Panchamalai. States: Odisha, Telangana, AP, Karnataka, Tamil Nadu. Eastern and Western Ghats meet at the Nilgiri Hills.

Differences: Western Ghats vs Eastern Ghats

Aspect	Western Ghats	Eastern Ghats
Direction	N–S along western coast (Tapi → Kanyakumari)	NE–SW parallel to eastern coast (Odisha → Kaveri)
Avg. Height	1,500 m	~600 m (considerably lower)
Avg. Width	50–80 km	100–200 km (wider)
Continuity	Continuous; crossed only via passes	Discontinuous; eroded by large rivers into separate parts
Structure	Complete unit	Lacks uniformity
River Origin	Source of major peninsular rivers	No major river originates here
Rainfall	Perpendicular to SW monsoon → heavy western slope rainfall	Parallel to SW monsoon → less summer rainfall

(ii) Central Highlands

Bounded by: Aravalli (W), vast plains (N), Deccan plateau (S), Rajmahal Hills in east (south of which lies Chotanagpur). Height: 700–1,000 m (decreases NE). Major ranges: Aravalli, Satpura, Vindhya, Chotanagpur. Also includes Malwa Plateau, Bundelkhand, Baghelkhand.

Aravalli Range

Meaning: 'Line of Peaks' (शिखरों की रेखा). **World's oldest folded mountain; now a residual mountain.** Extends from Majnu ka Tila (Delhi, NE) to Palanpur (Gujarat, SW) via Haryana and Rajasthan — ~692 km. ~80% in Rajasthan. Avg. height: ~930 m (increases southward).

(Highest peak: Gurushikhar 1722 m)(BPSC PYQ), Mount Abu, Sirohi district. Rivers: Luni, Banas, Sabarmati originate here.

Malwa Plateau

Meaning: 'Part of abode of Lakshmi' (Sanskrit: Malva). Spread across W. MP and SE Rajasthan. Bounded by Bundelkhand (N/E), Vindhya (S), Aravalli (W). Avg. height: 500–600 m; formed from basaltic lava. Rivers: Chambal, Banas, Shipra, Kali Sindh, Mahi. Black soil → good for agriculture.

Vindhya / Vindhya Range

Starts from Gujarat (W), extends through Rajasthan, MP, UP to Bihar (as Kaimur Hills) — total 1,086 km. Acts as watershed between rivers of northern plains and peninsular India. Diamond mine in Panna reserve (MP) (THE first river interlinking Ken- Betwa passes through this reserve). Major right-bank Yamuna tributaries originate here: Chambal, Sindh, Betwa, Ken.

Satpura Range

Determines the northern boundary of the Deccan Plateau. Parallel to Vindhya. Avg. height: 600–900 m. Narmada Valley to its north; Tapi/Tapi Valley to its south. Both rivers flow in rift valleys; Satpura is a block mountain between them. Extends from E. Gujarat through Maharashtra, MP to Chhattisgarh.

Regional names: Rajpipla Hills (Gujarat), Mahadev Hills (MP), Maikal Range (MP & Chhattisgarh).

Highest peak: Dhoopgarh (1,350 m), Panchmarhi, Mahadev Hills. Amarkantak (1,067 m) on Amarkantak plateau — radial drainage pattern: Son (N), Narmada (W), Mahanadi (E).

Chotanagpur Plateau

Called the mineral storehouse of India. Spread over Jharkhand (max), Bihar, WB, Odisha, Chhattisgarh. Avg. height: 700 m. North: Ganges Plain; South: Mahanadi Basin. Sub-regions: Palamu, Koderma, Hazaribagh, Rajmahal Hills, Ranchi, Pat Plateau, Singhbhum. Source of Damodar, Barakar, N. Koel, S. Koel, Swarnarekha rivers. Highest peak: Sammed Shikhar / Parasnath (1,365 m), Giridih — major Jain pilgrimage site. Rich in iron ore, aluminium ore, uranium and coal.

(iii) North Eastern Plateau

Extended part of the Peninsular Plateau. Separated from the main plateau when the Indian plate's NE shift created a rift valley between Rajmahal Hills and Meghalaya Plateau (later filled by alluvium). Includes Meghalaya Plateau and Karbi Anglong Hills (Assam).

Meghalaya Plateau subdivisions (named after tribes): Garo Hills, Khasi Hills, Jaintia Hills (BPSC PYQ). Height increases W to E. Storehouse of coal, iron, limestone, uranium.

Mawsynram (East Khasi Hills) receives the highest average annual rainfall in the world.

Significance of the Peninsular Plateau

Aspect	Details
Mineral Wealth	Metallic: iron, manganese, copper, bauxite, chromium, gold, silver, zinc, lead, mercury. Non-metallic: coal, diamonds, mica, marble, limestone, asbestos.
Soils & Crops	Regur (black soil) for cotton, millets, maize, pulses, sugarcane, citrus. Red soil for tea, coffee, rubber, cashews, spices, tobacco, groundnuts.
Archaeology Soils	Archaeon, Dharwar, Cuddapah, Vindhyan rocks → red/brown/lateritic soils supporting groundnuts, cashews, pulses.
Forests	W. Ghats, E. Ghats, Nilgiris: tropical evergreen/deciduous forests; teak, sal, sandalwood, ebony, mahogany, bamboo, rosewood.

Hydroelectricity	E-flowing rivers create waterfalls, rapids, gorges; W. Ghats rivers also generate HEP. Dams, canals and reservoirs aid irrigation.
Tourism	Ooty, Kodaikanal, Mahabaleshwar, Khandala, Matheran, Pachmarhi, Mount Abu.
Medicinal Plants	W. and E. Ghats forests rich in high-quality medicinal herbs.
Tribal Culture	Many Scheduled Tribes inhabit the hill ranges. Dravidian culture dominates south of Vindhyas.

INDIAN DESERT

A desert has avg. annual rainfall < 25 cm, very low soil moisture and sparse vegetation. Evaporation rate > rainfall rate. Indian deserts are divided by temperature:

- Hot Desert — Thar Desert
- Cold Desert — Ladakh Desert

Thar Desert

7th largest hot desert in the world; highest population density of any desert globally. **Location:** West and NW of Aravalli range. Avg. annual rainfall < 150 mm (15 cm). Covers ~61% of Rajasthan's area.

Geology: Believed to be part of the sea in the Mesozoic period (confirmed by Wood Fossil Park, Akal, Jaisalmer). Geologically an extension of the Peninsular Plateau, but surface shaped by physical weathering and wind erosion.

Landforms: Shifting sand dunes, Yardangs, mushroom rocks, Playas, Barkhans, Oases.

Drainage: Mostly seasonal. Luni is the main river. (UPSC PYQ) Some rivers have inland drainage (disappear into playa lakes). Playa lakes: Lunkarsar (Bikaner), Didwana (Nagaur), Sambhar (Jaipur) — saline water → salt production.

Cold Desert (Ladakh)

In the rain shadow of the Himalayas, extending from Karakoram (N) to the Himalayas (S). Avg. annual rainfall < 10 cm. Altitude: 3,000–5,000 m → very cold climate. Snowstorms, snowfall and landslides common. Soil is not very fertile; minimal water resources.

Seabuckthorn — Leh berry / Ladakh Gold; a superfood native to Ladakh.

COASTAL PLAIN

Divided by location and geomorphological processes into Western Coastal Plain and Eastern Coastal Plain. Both originated through river deposition and sea-wave erosion/deposition.

Western Coastal Plain

Type: Submerged coastal plain → narrow strip → favours natural ports.

Natural ports: Kandla, Wadhawan, Mumbai, JLNehru (Nhava Sheva), Mazgaon, Marmagaon, Mangalore, Cochin.

Divisions (N to S): Kutch & Kathiawar coast (Gujarat), Konkan coast (Maharashtra), Goa coast, Karnataka Plain (Goa → Mangalore), Malabar coast (Kerala).

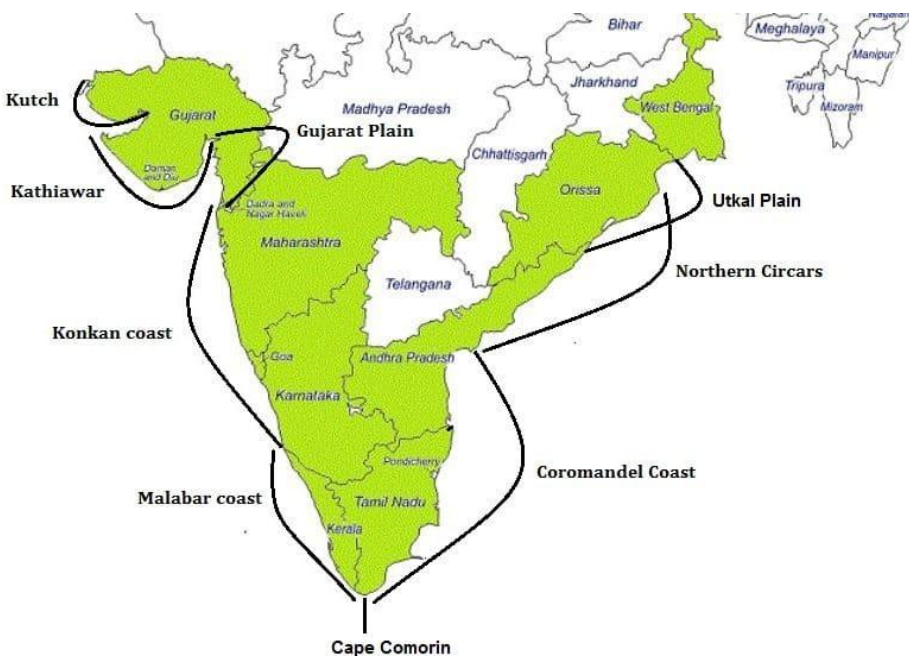
Narrow in the middle; wide in north and south. Rivers form estuaries, not deltas.

Malabar coast special

topography: Kayal (backwaters)

— used for fish breeding, fishing,

inland boating and tourism. Famous Nehru Trophy Valamkali (boat race) held at Punnamada Kayal, Kerala.



Eastern Coastal Plain

Type: Elevated (emergent) coast → wider than western plain; fewer natural ports. Continental shelf width: 500 km → port development difficult.

Main ports: Paradip, Visakhapatnam, Chennai, Ennore, Tuticorin.

Major deltas: Mahanadi, Godavari, Krishna, Kaveri.

Feature	Location	Significance
Northern Circar	North of Godavari river	Odisha & N. Andhra Pradesh coast
Coromandel Coast	Tamil Nadu coastline	S. Eastern coastal plain
Chilika Lake	Odisha coast	Country's largest saltwater lagoon (Ramsar wetland #1)
Kolleru Lake	Andhra Pradesh	Between Godavari & Krishna deltas
Pulicat Lake	Nellore district, AP (near AP–TN border)	Near Sriharikota (Satish Dhawan Space Centre)

ISLANDS

India has two major island groups — one in the Arabian Sea and one in the Bay of Bengal.

Andaman & Nicobar Islands (Bay of Bengal)

About 572 islands; located between 6°–14° N latitude and 92°–94° E longitude. Two major island groups: Ritchie Islands and Labyrinth Islands. Divided into Andaman (north) and Nicobar (south).

Feature	Detail	Location / Note
Capital	Port Blair (Sri Vijayapuram)	South Andaman
Active Volcano	Barren Island	Only active volcano in India (Andaman)
Dormant Volcano	Narcondam Island	Andaman Sea
Southernmost Point of India	Indira Point (Pygmalion Point)	Great Nicobar Island
Separation	10° Channel	Little Andaman — Car Nicobar
Duncan Passage	Between Rutland Island and Little Andaman	—

Highest Peaks — Andaman & Nicobar

- Saddle Peak (738 m) — North Andaman
- Mount Diavoli (515 m) — Middle Andaman
- Mount Koyob (460 m) — South Andaman
- Mount Thuiller (642 m) — Great Nicobar

Renamed Islands: Ross Island → Netaji Subhash Chandra Bose Island; Neil Island → Shaheed Island; Havelock → Swaraj Island.

Tribes: Jarawa, Ongej, Nicobari, Sentinelese (population continuously declining).

Main crops: Rice; cash crops: coconut, betel nut, pineapple; also banana, papaya, mango, guava.

Lakshadweep & Minicoy (Arabian Sea)

Located between 8°–14° N latitude and 71°–74° E longitude. Distance: 280–480 km from Kerala coast. Entire archipelago made of coral deposits (Atoll). **36 islands total**; 11 inhabited.

Feature	Detail	Note
11° Channel	Divides Lakshadweep into: Amindivi Islands (N) and Kannanore Islands (S)	—
9° Channel	Between Lakshadweep and Minicoy	—
8° Channel	Between Minicoy and Maldives	Separates Minicoy from Maldives
Largest Island	Andrott (4.9 sq. km)	Lakshadweep group

Second Largest / Southernmost	Minicoy Island	Also separated by 8° channel
Capital	Kavaratti	Lakshadweep & Minicoy
Main Cash Crop	Coconut	Highest coconut productivity in India

RIVER DRAINAGE PATTERN

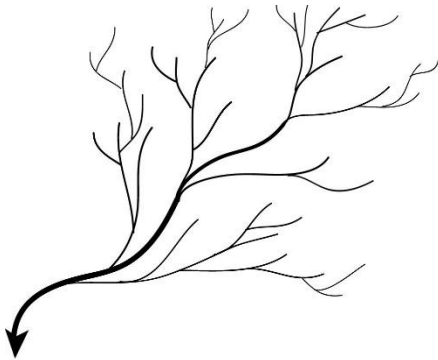
Definition: The pattern formed by the main river and its tributaries from the source to its mouth is called the river drainage pattern. It depends on factors such as dominance of hard or soft rocks, slope, and velocity of water.

DENDRITIC DRAINAGE PATTERN

Description: Tributaries of the main stream join in a way that the geometric pattern formed resembles the branches of a tree. Develops on flat or gently sloping land where rock resistance is uniform.

Examples: Ganga, Brahmaputra (North Indian plains); Godavari, Mahanadi (peninsular rivers).

Dendritic Pattern — Diagram



Pattern resembles a tree's branching structure. No dominant structural control.

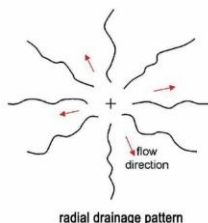
RADIAL DRAINAGE PATTERN (CENTRIFUGAL)

Description: Rivers originate from a central mountain or hill and flow outward in all directions — like spokes of a wheel. Also called the **centrifugal drainage pattern**.

Examples:

- **Amarkantak Plateau:** Source of Narmada (flows west), Mahanadi (flows east), Son (flows north) — classic radial pattern.
- **Also:** Rajmahal Hills, Girnar (Gujarat), Mikir Hills (Assam).

Radial Pattern — Diagram



Central hill / plateau (e.g. Amarkantak)

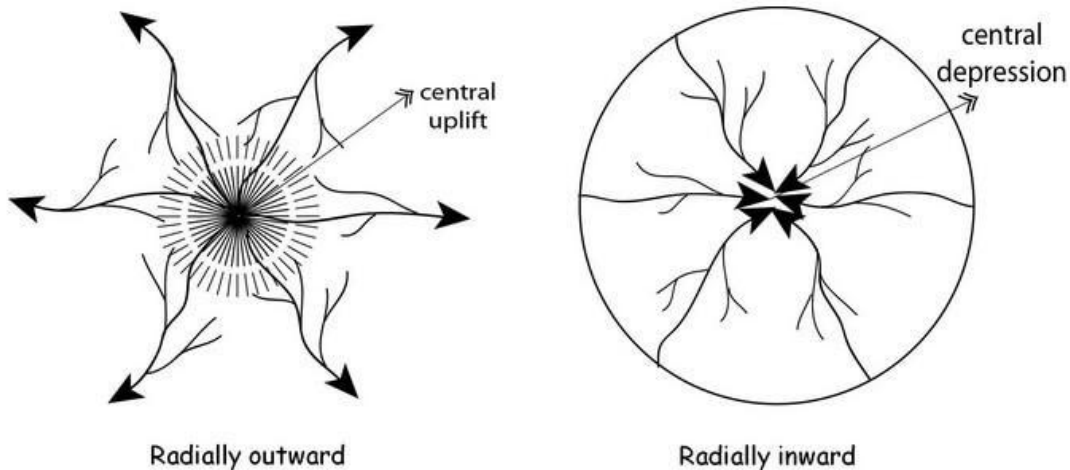
Rivers flow outward in all directions.

CENTRIPETAL DRAINAGE PATTERN

Description: Rivers flow inward from all directions and converge into a central lake, pond or depression. This is the reverse of the radial pattern.

Example: Loktak Lake (largest natural freshwater lake of North eastern India with floating vegetation-PHUMDIS), Manipur — rivers from surrounding hills drain into this central lake.

Centripetal Pattern — Diagram

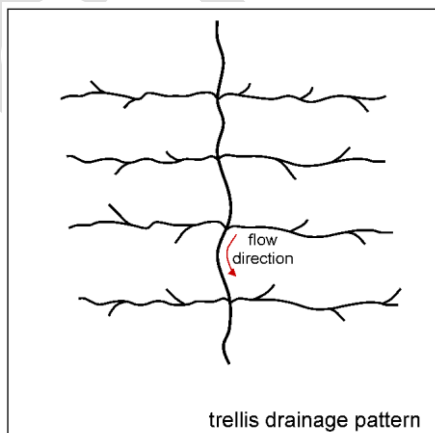


TRELLIS DRAINAGE PATTERN

Description: Primary tributaries of the main river flow parallel to each other, while secondary tributaries join primary tributaries at right angles (90°). Resembles the pattern of a trellis or garden grid.

Example: Rivers of the Chotanagpur Plateau.

Trellis Pattern — Diagram



RECTANGULAR DRAINAGE PATTERN

Description: Develops where rock joints and fractures are arranged in a rectangular pattern. Tributary rivers meet the main river at right angles (90°) through the same joints and junctions. Similar to trellis but controlled by two sets of rock joints at right angles.

Examples: Rivers of the Vindhya region; Krishna River (upper reaches); Kosi River (upper reaches).

Rectangular Pattern — Diagram

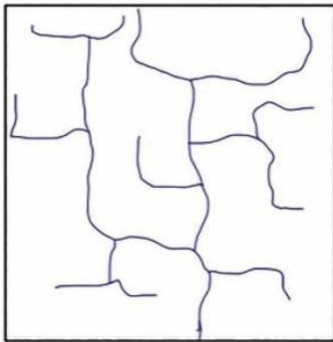


Fig.- Rectangular Pattern

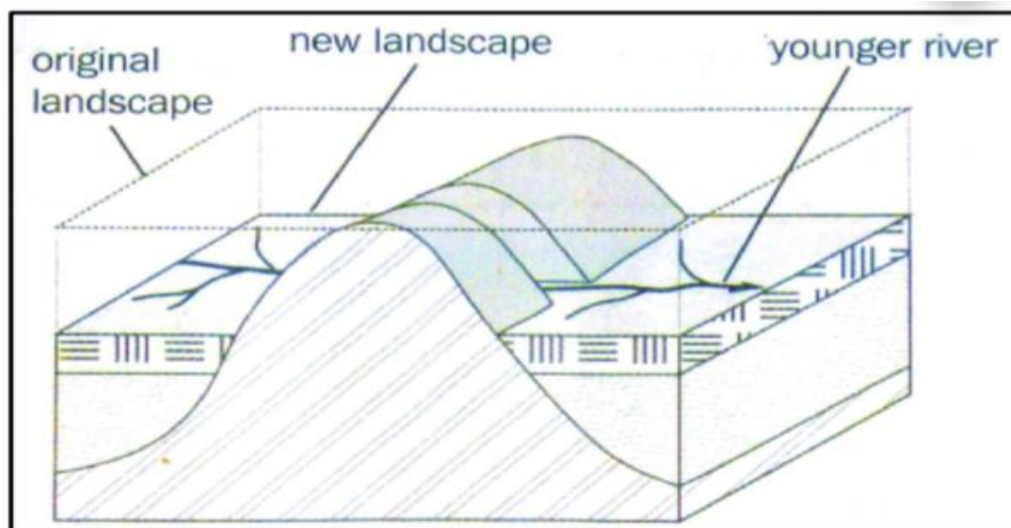
SUPERIMPOSED DRAINAGE PATTERN

Description: Due to continuous river erosion, the overlying soft rocks are worn away and the river begins to flow freely on the underlying hard rocks. The river was originally established on the soft rocks above, but has been 'let down' onto the harder rocks beneath — hence 'superimposed'.

Examples: Damodar, Swarnarekha, Chambal, Banas rivers.

Superimposed Pattern — Diagram

Superimposed Drainage Patterns



OVERVIEW: MAJOR DRAINAGE PATTERNS

Pattern	Key Characteristic	Examples
Dendritic	Tributaries resemble branches of a tree; flat/gentle slope terrain	Ganga, Brahmaputra (North Indian plains); Godavari, Mahanadi (peninsular)
Radial / Centrifugal	Rivers flow outward from a central hill/mountain in all directions	Amarkantak Plateau (source of Narmada, Mahanadi, Son); Rajmahal Hills; Girnar; Mikir Hills
Centripetal	Rivers flow inward from all directions into a central lake/depression	Loktak Lake, Manipur
Trellis	Primary tributaries flow parallel; secondary tributaries join at right angles	Rivers of Chotanagpur Plateau
Rectangular	Tributaries meet main river at right angles through rectangular rock joints	Vindhya region rivers; Krishna (upper reaches); Kosi (upper reaches)
Superimposed	Continuous erosion exposes hard rocks; river flows freely on them	Damodar, Swarnarekha, Chambal, Banas

DRAINAGE SYSTEM OF INDIA**BASIC CONCEPTS**Drainage System

The flow of water through certain channels or streams is called 'drainage'. The network of these channels is called the 'drainage system'. The drainage system of an area is the result of geological time period, nature and structure of rocks, topography, slope and amount of flowing water.

Term	Definition
Catchment Area	The area of land from which water flows into a river. It is relatively small and a sub-unit of a river basin.
River Basin	Entire land area drained by a river and its tributaries. It is relatively large, encompassing more than one catchment area.
Watershed	The catchment area of small rivers. Also, the boundary separating one drainage basin from another is called a Water Divide.

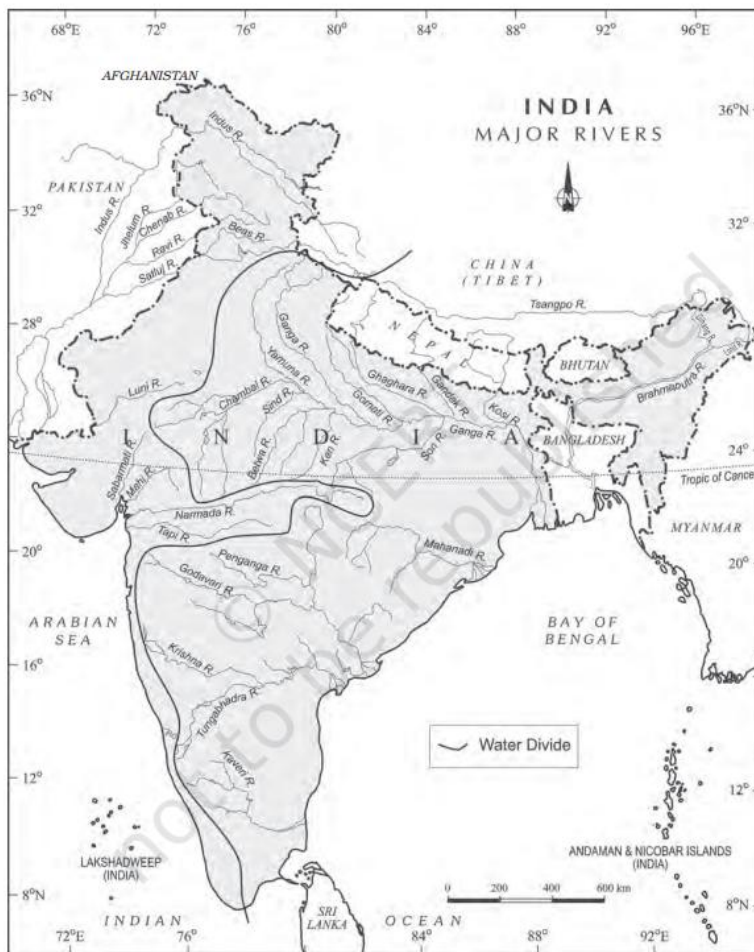


Figure 3.2 : Major Rivers of India

The drainage system of India is separated by the Delhi Ridge, Aravalli Range, Vindhyan and Satpura Ranges, and the Western Ghats. These ranges act as natural watersheds. On the basis of origin, the Indian drainage system is divided into two categories:

- Himalayan Drainage System
- Peninsular Drainage System

HIMALAYAN DRAINAGE SYSTEM

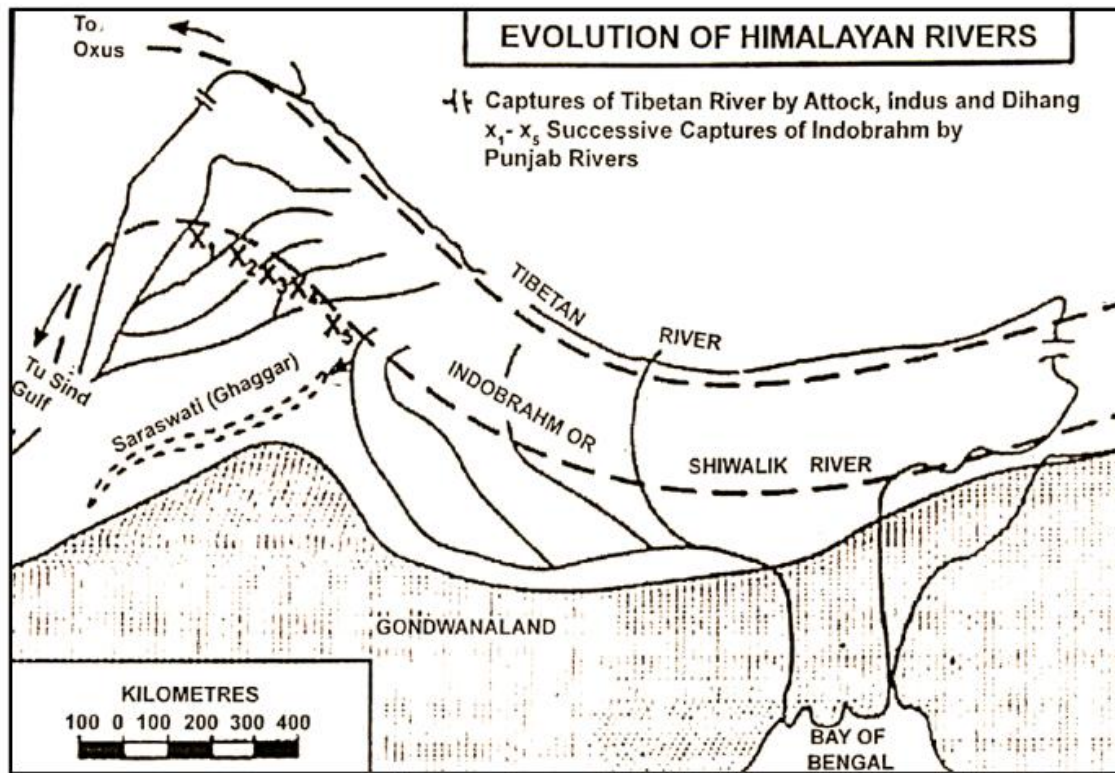
Development of Himalayan Drainage

The Himalayan drainage system has developed over a long geological period. Geologists believe that in the Miocene period, a massive river called Shivalik or Indo-Brahma flowed from Assam to Punjab along the entire longitudinal extension of the Himalayas, finally discharging into the Gulf of Sindh.

Over time, the Indo-Brahma river divided into three main drainage systems:

- Indus and its five tributaries (west)
- Ganga and its tributaries originating from the Himalayas (middle)
- Brahmaputra and its tributaries originating from the Himalayas (east)

This division occurred due to the upliftment of the Potwar Plateau during the Pleistocene period, which became the watershed between the Indus and Ganga systems. The Malda Gap, formed during middle Pleistocene between the Rajmahal Hills and Meghalaya Plateau, caused the Ganga and Brahmaputra to flow towards the Bay of Bengal.



Indus River System

Total Length	2,880 km (1,114 km in India)
Origin	Bokhar Chu glacier, Kailash Range, Tibet (altitude: 4,164 m)
Tibetan Name	Singi Khamban / Shermukh
Entry into India	Damchok, Ladakh
Final outfall	Arabian Sea (east of Karachi)

Right bank tributaries: Shyok (largest), Nubra, Shigar, Hunza, Gilgit

Left bank tributaries: Jhelum, Chenab, Ravi, Beas, Sutlej, Zaskar

Hydroelectric projects (Ladakh): Nimbo Bajgo, Dumkhar

Jhelum River

Originates from Verinag waterfall (Sheshnag Lake) in the foothills of Pir Panjal range, south-east of Kashmir valley. Flows through Srinagar and Wullar Lake. After Muzaffarabad, forms a 170 km border between India and Pakistan. Meets Chenab near Jhang in Pakistan.

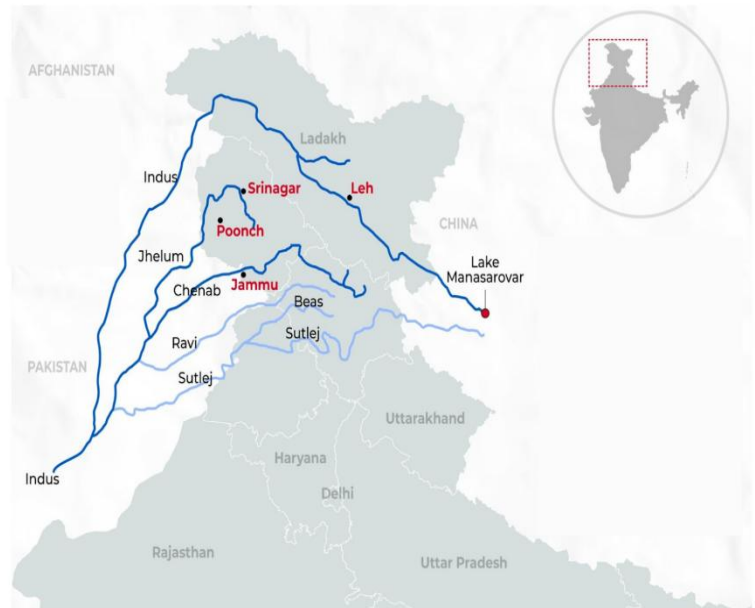
Major tributary: Kishanganga

Projects: Tulbul Project, Uri Hydroelectric Project

Chenab River

The largest tributary of Indus. Formed by the confluence of Chandra (originates from Chandra Tal, Himachal Pradesh) and Bhaga (originates from Surya Tal near Bada Lacha Pass, Himachal Pradesh).

Projects in Kashmir: Baglihar, Salal, Dulhasti

Ravi River

Originates west of Rohtang Pass in Kullu Hills of Himachal Pradesh. Flows through Chamba Valley. Joins Chenab near Multan in Pakistan.

Beas River

Originates from Beas Kund near Rohtang Pass, Himachal Pradesh. Meets Sutlej at Harike in Punjab. City of Manali is on its right bank.

Notable: Harike Barrage feeds the Indira Gandhi Canal — world's largest canal irrigation project

Project: Pong Hydroelectric Project (Kangra district)

Sutlej River

An antecedent river. Originates from Rakas/Rakshas Tal near Mansarovar Lake, Tibet (known as Logchen Khebab). Total length 1,450 km. Enters India through Shipki La Pass into Kinnaur district.

Projects: Bhakra-Nangal Project (Bilaspur), Nathpa Jhakri Project (Himachal Pradesh)

BCW BITS INDUS RIVER SYSTEM ALONG WITH ITS TRIBUTARIES IS AN IMPORTANT THEME OF BPSC AND UPSC

Ganga River System

Total Length	2,525 km
Origin	Gangotri Glacier near Gomukh, Uttarkashi, Uttarakhand (altitude: 3,900 m)
Original name	Bhagirathi (until Devprayag)
Course	Uttarakhand → Uttar Pradesh → Bihar → Jharkhand → West Bengal
Basin area (India)	~8.6 lakh sq. km (largest drainage system in India)



Panchprayag — Five Holy Confluences

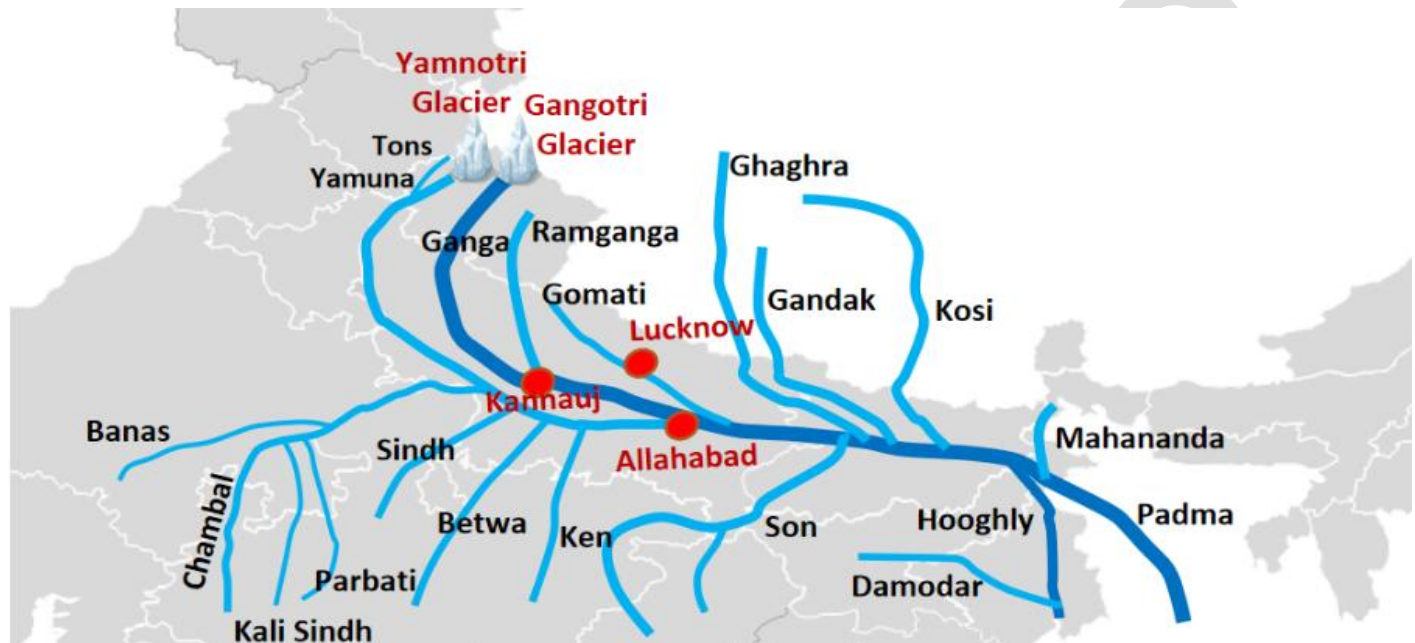
Confluence	Prayag
Alaknanda + Dhauliganga	Vishnuprayag (Joshimath)
Alaknanda + Nandakini	Nandprayag
Alaknanda + Pindar	Karnaprayag
Alaknanda + Mandakini	Rudraprayag
Alaknanda + Bhagirathi	Devprayag



At Farakka (West Bengal), the Ganga divides into two distributaries: Hooghly (flows into Bay of Bengal at Gangasagar) and the main stream (flows into Bangladesh as Padma, joins Jamuna and Meghna, and flows into the Bay of Bengal). The Ganga and Brahmaputra together form the world's largest delta — the Sundarbans Delta (over 10,000 sq. km).

Left Bank Tributaries of Ganga (West to East) (UPSC PYQ)

Ramganga | Gomti | Ghaghra | Gandak | Budhi Gandak | Kosi | Mahananda



Ramganga

Originates from Namik glacier, Pithoragarh (Kumaon Himalayas, Uttarakhand). Flows through Almora and Jim Corbett National Park. Joins Ganga at Kannauj district, UP.

Gomti (960 km)

Originates from Gomti/Fulhar Lake, Pilibhit district, UP. Only left bank tributary of Ganga in UP with a plains source. Flows through Lucknow, Sultanpur, Jaunpur. Joins Ganga at Ghazipur. Major tributary: Sai River.

Sharda (350 km)

Originates from Milam Glacier, Pithoragarh (known as Gauri Ganga). Called Kali River on India-Nepal border. Becomes Sharda in Pilibhit, UP. Joins Ghaghra at Brahmaghat, Bahraich

LEFT BANK TRIBUTARY OF RIVER GANGA IN BIHAR

Ghaghra (1,080 km)

Largest tributary of Ganga by water volume. Originates from Mapchachungo glacier, south of Kailash Mansarovar, Tibet. Known as Mapcha Sangpo (Tibet), Karnali (Nepal), Saryu (Ayodhya). Enters Bihar at Guthni, Siwan. Joins Ganga at Revelganj, Saran. Tributaries: Sharda, Rapti, Chhoti Gandaki, Doha.

Gandak (260 km)

Formed by confluence of Kali Gandak and Trishul Ganga. Originates between Dhaulagiri and Everest in Nepal. Known as Narayani in Nepal. Enters Bihar at Bagaha, West Champaran. Joins Ganga near Hajipur and Sonpur.

Budhi Gandak (320 km)

Originates from Chautarwa Chaur, Bisambarpur, West Champaran. Only left bank Ganga tributary in Bihar with plains source. Flows through Darbhanga, Samastipur, Begusarai. Joins Ganga in Khagaria. Major tributary: Bagmati.

Bagmati (394 km)

Originates from Baghdwar, Shivapuri hills, Nepal Himalayas. Flows through Kathmandu (Nepal's most sacred river). Enters Bihar at Dheng, Sitamarhi. Major tributary: Kamla-Balan.

Kamla-Balan (120 km)

Kamala originates from Sindhuliagarhi, Mahabharata Range, Nepal; enters Bihar at Jaynagar, Madhubani. Balan originates from Nepal Himalayas; enters Bihar at Jhanjharpur, Madhubani. Combined stream joins Bagmati at Badlaghat, Khagaria.

Kosi (260 km)

Known as the 'Sorrow of Bihar' due to frequent course changes and floods. Source is north of Mt. Everest in Tibet (main stream: Arun Kosi). Called Saptakosi in Nepal. Enters Bihar at Supaul. Joins Ganga at Kursela Ghat, Katihar.

BCW Bits – The Kosi -Mehi (tributary of Mahananda river) is the second river interlinking project of INDIA, first was ken-betwa which is already mentioned in your notes

Mahananda (360 km)

Originates from Darjeeling Himalayas, West Bengal. Flows through Kishanganj, Purnia, Katihar districts. of Bihar. Last (easternmost) river to join Ganga's left bank (in Malda district, West Bengal)

Right Bank Tributaries of Ganga (West to East)

Yamuna | Karmanasha | Punpun | Falgu | Kiul | Chandan | Mayurakshi | Ajay | Damodar

Yamuna (1,376 km)

Longest tributary of Ganga; longest river in India not draining directly to sea. Originates from Yamunotri glacier, western slope of Bandarpooch range (altitude: 6,387 m), Uttarkashi. Flows through Uttarakhand, Himachal Pradesh, Haryana, Delhi. Meets Ganga at Prayagraj.

Left bank tributaries: Tons, Hindon, Rind, Varuna, Senger.

Right bank tributaries: Chambal, Sindh, Betwa, Ken (from peninsular plateau).

Karmanasha (76 km)

Originates from Kaimur Hills. Forms border between UP and Bihar. Joins Ganga near Chausa in Buxar district — westernmost right bank tributary of Ganga in Bihar.

Son (784 km)

Originates from Amarkantak plateau, Maikal range, Chhattisgarh. Second largest right bank tributary of Ganga (after Yamuna); largest in Bihar. Enters Rohtas district, Bihar. Joins Ganga near Maner, Patna district. Major tributaries: Northern Koel, Kanhar, Rihand, Gopad. Projects: Bansagar Dam (MP), Rihand Dam (UP), Indrapuri Barrage (Bihar).

Punpun (235 km)

Originates from Palamu district, Jharkhand (Chota Nagpur plateau). Flows through Aurangabad, Gaya, Jehanabad. Joins Ganga at Fatuha, Patna.

Falgu (135 km)

Formed by confluence of Lilajan (Niranjana) and Mohana rivers at Bodh Gaya. Buddha attained enlightenment on the banks of Niranjana. Joins Ganga via Mokama Tal, Patna.

Damodar (592 km)

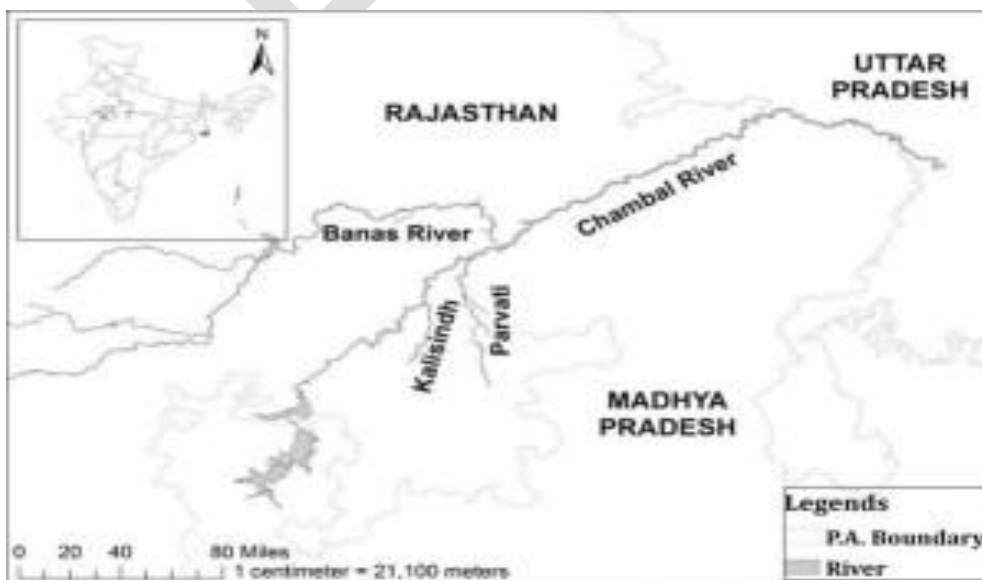
Once called the 'Sorrow of Bengal'. Originates from Chandwa village, Palamu, Jharkhand (Chotanagpur plateau). Flows through a rift valley in upper course. Joins Hooghly at Falta, South 24 Parganas. Major tributaries: Bokaro, Konar, Barakar.

Damodar Valley Project (1948) — India's first multipurpose river valley project (modeled on Tennessee Valley Project, USA). 8 dams total; prominent: Tilaiya Dam (Barakar), Maithon Dam (Barakar), Panchet Dam (Damodar), Konar Dam (Konar). Joint project of Jharkhand and West Bengal. **(BPSC PYQ)**

Major Right Bank Tributaries of YamunaChambal (965 km)

Originates from Janapau hill near Mhow, Indore district, MP (Vindhyan range). Flows through great ravines to Kota, Rajasthan. Meets Yamuna in Etawah district, UP. Famous for Chambal ravines. Major tributaries: Banas (largest — originates from Aravalli near Kumbhalgarh, Rajsamand), Kshipra, Kali Sindh, Parvati.

Chambal River Project (MP-Rajasthan joint): Gandhi Sagar Dam (Mandsaur, MP), Rana Pratap Sagar Dam (Chittaurgarh, Rajasthan), Jawahar Sagar Dam and Kota Barrage (Kota).



Sindh River

Originates from Vidisha district, MP (Malwa Plateau). Flows north-northeast. Joins right bank of Yamuna in Auraiya district, UP.

Betwa River

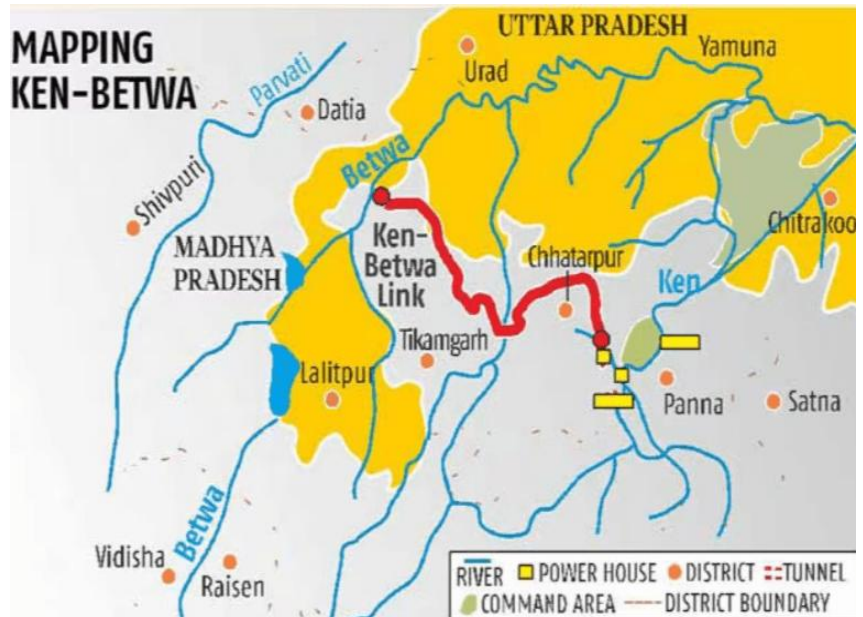
Originates from Raisen district, MP (Vindhyan range). Flows through Vidisha, Sanchi, Orchha, Lalitpur, Jhansi. Joins Yamuna at Hamirpur district. Largest river of Bundelkhand region.

Ken-Betwa Link Project — India's first national river linking project. Aims to carry Ken's excess water to Betwa to address Bundelkhand drought.

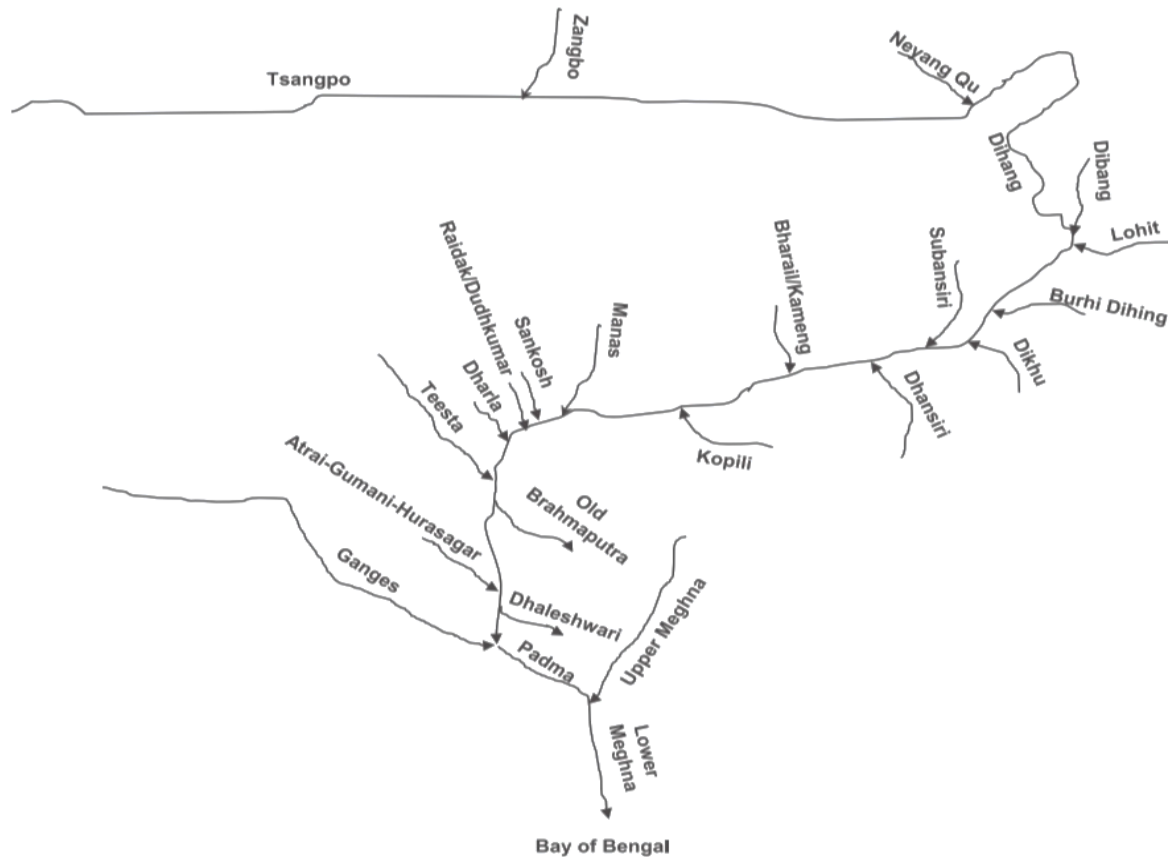
Major dams: Mata Tila Dam (Lalitpur), Parichha Dam (Jhansi), Samrat Ashok Sagar Dam (Vidisha-Raisen).

Ken River

Originates from Katni district, MP. Flows through Bundelkhand (MP-UP). Joins right bank of Yamuna in Banda district, UP. Flows through Panna National Park.

Brahmaputra River System

Total Length	916 km (in India)
Origin	Chemayungdung glacier, near Mansarovar Lake, Kailash Range
Names across journey (MOST ASKED BPSC THEME)	Sangpo (Tibet) → Siang/Dihang (near Namcha Barwa) → Brahmaputra (India) → Jamuna (Bangladesh)
Entry into India	Sadiya, Arunachal Pradesh (takes a 'U' turn near Namcha Barwa)



From its origin, the Brahmaputra flows about 1,200 km eastward in the arid plains of southern Tibet as Sangpo (meaning 'purifier'). Near Namcha Barwa at the eastern end of the Himalayas, it forms a deep gorge and emerges as Siang/Dihang. At Sadiya, it is joined by Dibang and Lohit rivers and becomes the Brahmaputra. After a 750 km journey through the Assam valley, it turns south, enters Bangladesh (joining Teesta), and becomes Jamuna. Jamuna meets Padma, then Meghna, and flows into the Bay of Bengal.

Left bank tributaries: Dibang, Lohit, Burhi Dihing, Disang, Dhansiri, Kopili

Right bank tributaries: Subansiri, Kameng, Manas, Sankosh, Raidak, Teesta

MAJULI: World's largest river island, formed by the Brahmaputra in Assam. Area: 880 sq. km. Given district status in 2016.

PENINSULAR DRAINAGE SYSTEM

Evolution of Peninsular Drainage

Three major geological events shaped peninsular India's drainage:

- During Early Tertiary period, the western flank of the peninsula subsided below sea level, disturbing the symmetrical plan of rivers.
- Himalayan upliftment caused the northern peninsular block to subside, forming fault basins. Narmada and Tapi flow in these fault valleys — hence they lack alluvial and delta deposits.
- The peninsular section tilted from NW to SE, orienting drainage towards the Bay of Bengal.

Rivers Draining into the Bay of Bengal

Swarnarekha / Subarnarekha

Originates from Nagdi village, south of Ranchi district, Chota Nagpur plateau. Flows through Jharkhand (through Jamshedpur), West Bengal, Odisha. Forms Hundru Falls in Ranchi. Major tributaries: Radhu, Kanchi, Kharkai. Notable: Only river in Jharkhand that flows directly into the Bay of Bengal without joining another river.

Brahmani River

Formed by confluence of Shankha and South Koel rivers near Rourkela, Odisha. Second largest river in Odisha (after Mahanadi). Forms large delta joining Baitarani/Dhamra river. **Bhitarkanika National Park (Kendrapara district) — second largest single-patch mangrove forest in India; Ramsar wetland;** famous for brackish water crocodiles. Gahirmatha Marine Sanctuary (world's largest Olive Ridley turtle nesting site) lies between the mouths of Dhamra and Brahmani rivers. Nesting season: November to April.

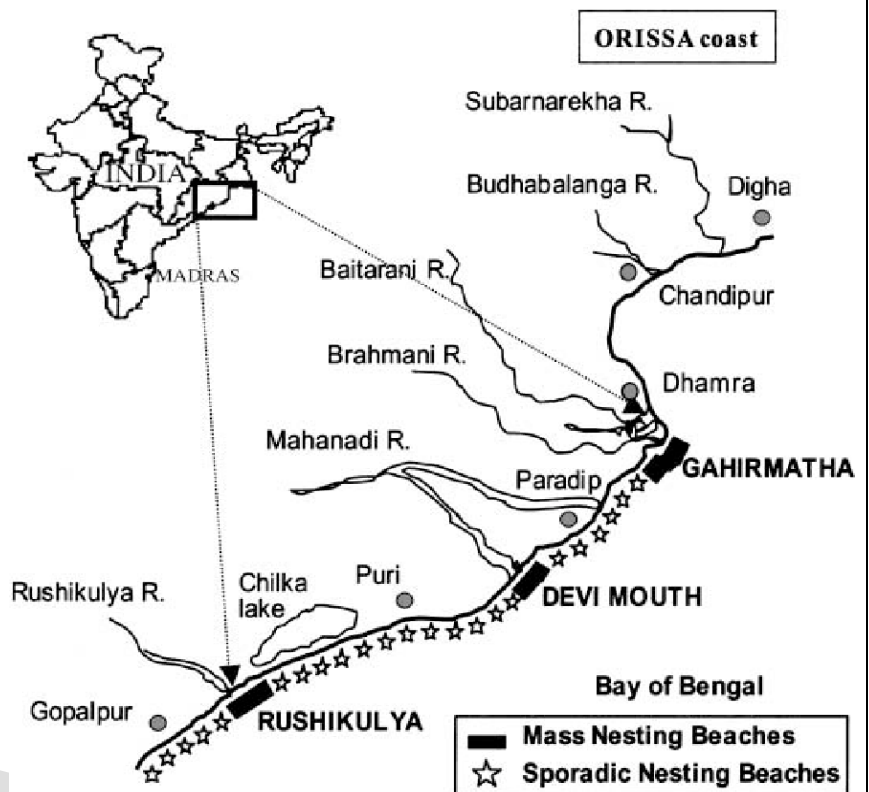
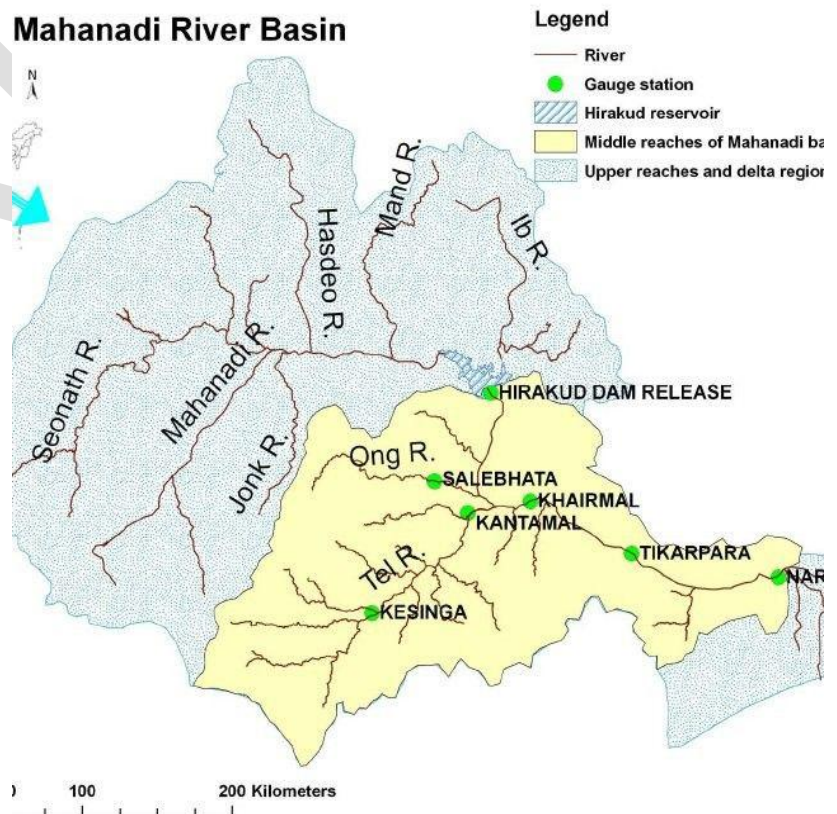
Mahanadi (890 km)

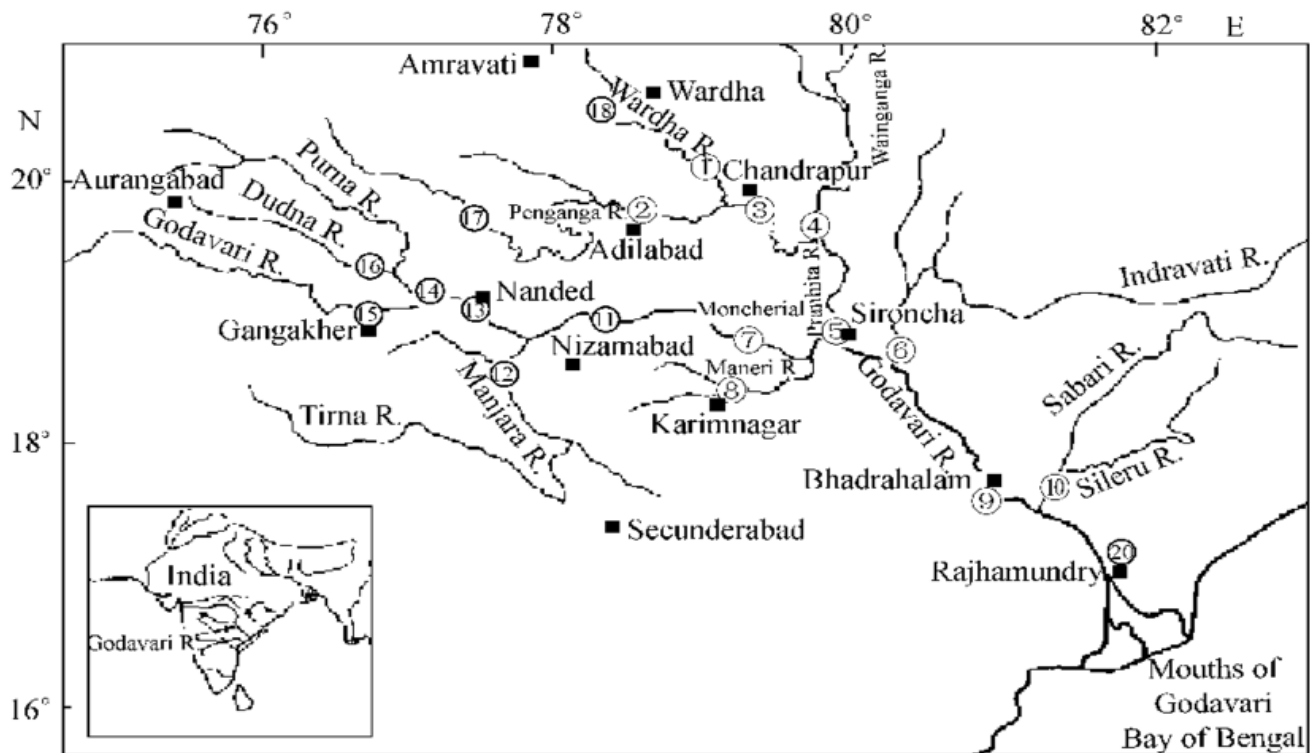
Originates from Sihawa hill, Dhamtari district, Chhattisgarh (Amkantak plateau). Divides near Cuttack forming a large delta. 53% basin in MP+Chhattisgarh; 47% in Odisha. Left bank tributaries: Shivnath, Hasdeo, Mand, Ib. Right bank tributaries: Ong, Tel, Jonk.

Hirakund Dam (1957) on Mahanadi near Sambalpur — world's longest earthen dam (25.79 km). Hirakund Lake (743 sq. km) — largest artificial lake in Asia; declared Ramsar wetland in 2021.

Rushikulya River

Originates from South Udayagiri Hills, Kandhamal district, Odisha. Flows into Bay of Bengal in Ganjam district. River mouth is a famous Olive Ridley turtle nesting site.

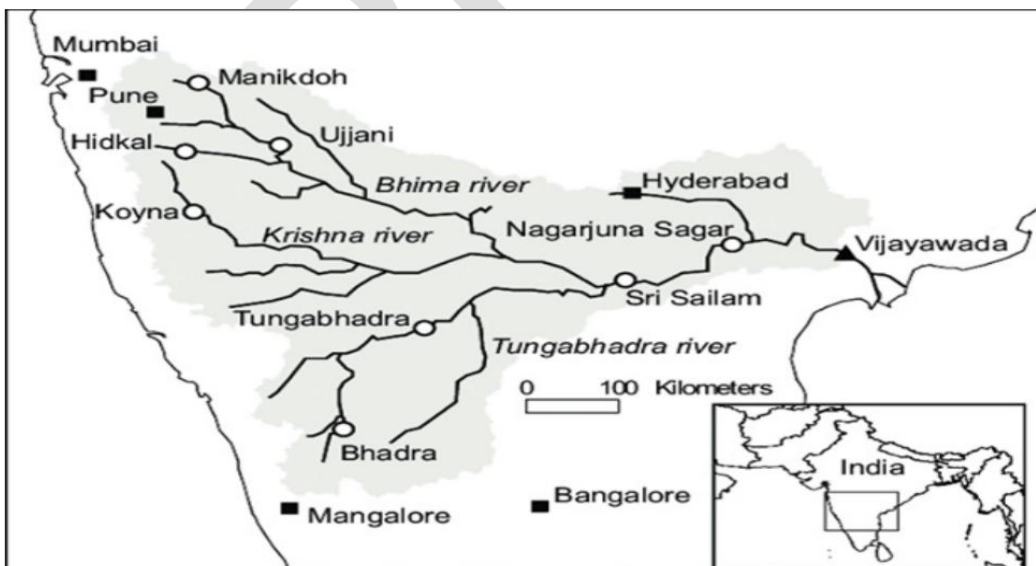
**Mahanadi River Basin**

Godavari River (1,465 km)

Largest river of peninsular India; second largest river in India. Called Dakshin Ganga and Vriddha Ganga.(BPSC AND UPSC PYQ) Originates from Trayambak plateau, Nasik district, Maharashtra. Flows into Bay of Bengal. Tributaries span Maharashtra, MP, Chhattisgarh, Odisha, Telangana, Andhra Pradesh (49% catchment in Maharashtra).

Left bank tributaries: Purna, Penganga, Wardha, Venganga, Pranahita, Indravati.

Right bank tributaries: Pravara, Manjira. Nizam Sagar Dam on Manjira.

Krishna River (1,400 km)

Second largest river of peninsular India.(UPSC PYQ) Originates from **Mahabaleshwar hill**, Western Ghats. Forms large delta near Vijayawada. 27% catchment in Maharashtra, 44% in Karnataka, 29% in Telangana and AP.

Left bank tributaries: Bhima, Musi (Hyderabad is on Musi). Right bank tributaries: Koyna, Panchganga, Dudhganga, Ghatprabha, Malaprabha, **Tungabhadra** (largest tributary). (Question on Tungabhadra already asked in UPSC 2023, so can be probable question in BPSC as well,)

BCW BITS: The Vijayanagara Empire was located along the banks of the Tungabhadra River (this location is currently known as Hampi).

Projects: Shivaji Sagar Dam (on Koyna), Nagarjuna Sagar Dam (on Krishna, in Nalgonda/Guntur) — largest masonry dam in India.

Kaveri River (800 km)



Originates from **Brahmagiri Hills**, Kogadu/Coorg district, Karnataka. Flows almost year-round due to dual monsoon — upper catchment receives SW monsoon (summer), lower catchment receives NE monsoon (winter). Basin: 3% in Kerala, 41% in Karnataka, 56% in Tamil Nadu.

Forms Shivasamudram Falls in Mandya district, Karnataka. Krishnaraja Sagar Dam (1931) at Mandya. Largest tributary: Hemavati (left bank). Right bank: Lakshmantirtha, Suvarnavati, Kabini, Bhavani, Amaravati. Mettur Dam / Stanley Reservoir (1934) in Salem district, Tamil Nadu.

BCW BITS: Inter-state river disputes in India are governed by **Article 262** of the Indian Constitution. This article gives Parliament the power to make laws to resolve water sharing disputes between states.

- **Examples:** The Krishna Water Disputes Tribunal and the Cauvery (Kaveri) Water Disputes Tribunal.

Rivers Draining into the Arabian Sea

Narmada River (1,312 km)

Originates from Narmada Kund, Amarkantak plateau, Maikal range, Annupur district, MP. Flows westward through a rift valley (between Satpura range in south and Vindhya range in north). Cuts marble rocks at Bhedaghat near Jabalpur — forms Dhuandhar Waterfall. Forms Asia's longest estuary (27 km) before meeting the Arabian Sea (Gulf of Khambhat), Bharuch district, Gujarat.

Projects: Sardar Sarovar Dam (Narmada district, Gujarat), Indira Sagar Dam and Omkareshwar Dam (Khandwa district, MP), Bargi Dam (Jabalpur), Maheshwar Dam (Khargone).

Tapi / Tapti River

Originates near Multai, Betul district, MP, south of Satpura range. Flows westward in a rift valley. Meets Arabian Sea (Gulf of Khambhat), Surat district, Gujarat. 79% drainage in Maharashtra, 15% in MP, 6% in Gujarat. Project: Ukai Dam (Surat district).

Sabarmati River

Originates from Udaipur district, Rajasthan (south of Aravalli range). Flows SW through Gujarat into Gulf of Khambhat. Gandhinagar (Gujarat's capital) and Ahmedabad are on its banks.

Mahi River

Originates from Dhar district, MP (northern slope of Vindhya range). Flows NW through Jhabua, Ratlam (MP), enters Banswara (Rajasthan), then turns south into Kheda district, Gujarat. Flows through three states (MP, Rajasthan, Gujarat) and **crosses the Tropic of Cancer twice**.

Luni River

Originates from western slope of Nag Mountain, Ajmer (Pushkar valley, Aravalli range) — initially called Sagarmati; becomes Luni at Govindgarh. Flows through Nagaur, Pali, Jodhpur, Balotra, Barmer, Jalore. Merges into Rann of Kutch, Gujarat. Water is sweet till Balotra; becomes saline thereafter due to high soil salinity.

Periyar River (244 km)

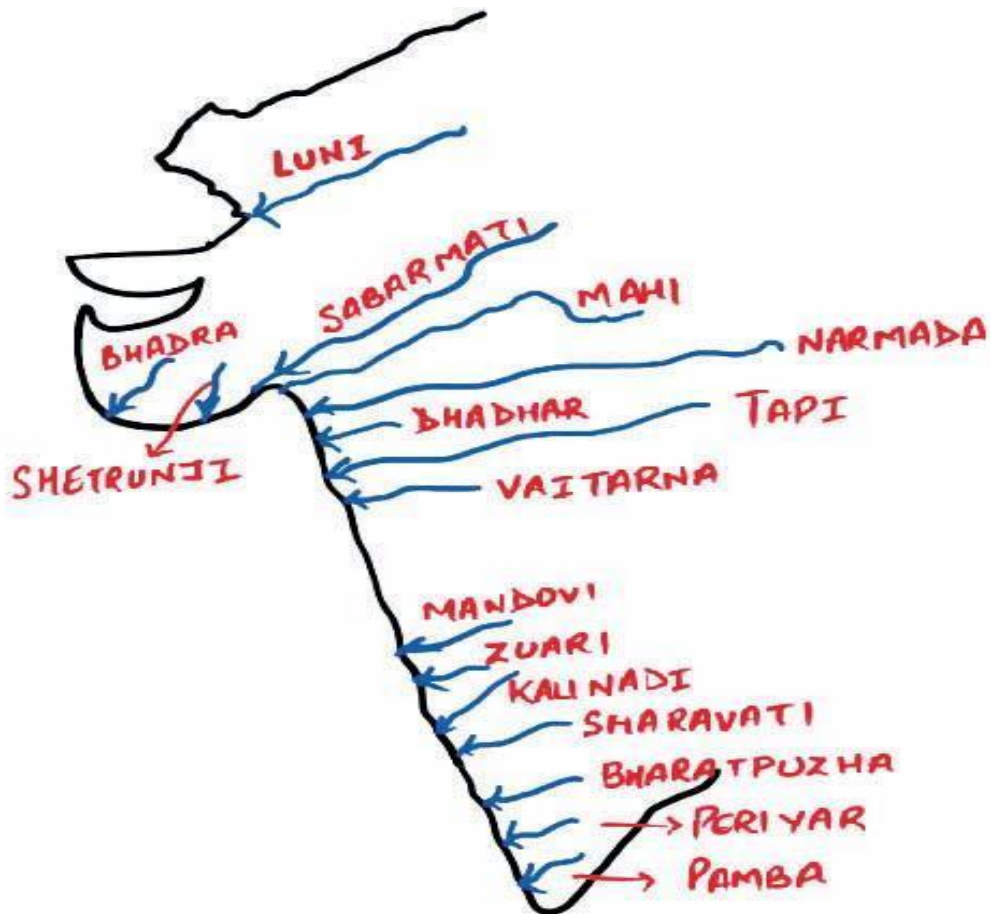
Originates from Sivagiri Hills, Western Ghats (Tamil Nadu). Lifeline of Kerala; longest river in Kerala. Flows through Periyar National Park (Periyar Tiger Reserve). Drains into Arabian Sea. Provides water supply to Kochi. Projects: Mullaperiyar Dam, Idukki Dam.

Zuari River (145 km)

Longest river in Goa. Originates from Western Ghats. Flows into Arabian Sea near Marmugao harbour. Port city of Vasco da Gama is at its mouth.

Mandavi / Mahadei River

Originates from Western Ghats, Karnataka. Flows 35 km in Karnataka, then 82 km through Goa. Empties into Arabian Sea at Panaji. Goa's capital Panaji is on its left bank. Famous Dudhsagar Falls is on this river. Mandavi and Zuari join at a common creek forming Mormugao harbour.



LONGEST RIVERS OF INDIA (Descending Order)

Rank	River	Length
1	Ganga	2,525 km
2	Godavari	1,465 km
3	Krishna	1,400 km
4	Yamuna	1,376 km
5	Narmada	1,312 km
6	Indus	2,880 km (1,114 km in India)
7	Ghaghra	1,080 km
8	Chambal	965 km
9	Gomti	960 km
10	Brahmaputra	916 km (in India)
11	Mahanadi	890 km
12	Kaveri	800 km
13	Son	784 km

MONSOON**INTRODUCTION**

Etymology: The word 'monsoon' is derived from the Arabic word 'mausim' meaning weather.

Definition: Winds that reverse direction with the change of seasons are called monsoon winds. In summer they blow from sea to land; in winter from land to sea.

India's Monsoon: India receives about **75% of its total average annual rainfall** during the southwest monsoon from June to September.

The nature of monsoon is uncertain — it affects vegetation, agricultural typology and social/cultural beliefs of India. Farmers must 'gamble' with it: some years see heavy floods, others severe drought. First studied by Arab geographer Al Masoodi; scientific study began in the colonial period.

THEORIES OF INDIAN MONSOON ORIGIN

Category	Theory	Key Proponent(s)
Classical Theories	Thermal Effect Theory	British climatologists; Dudley Stamp, Baker; Hadley's Thermal Cycle
Classical Theories	Equatorial Westerly Wind Theory	Flohn (meteorologist)
Modern Theories	Jet Stream Theory	Yest; Dr. P. Koteswaram (India)
Modern Theories	El Niño Theory	Gilbert Balker (1924); gained prominence after 1987 drought

Ideas before World War II = Classical; after WWII = Modern.

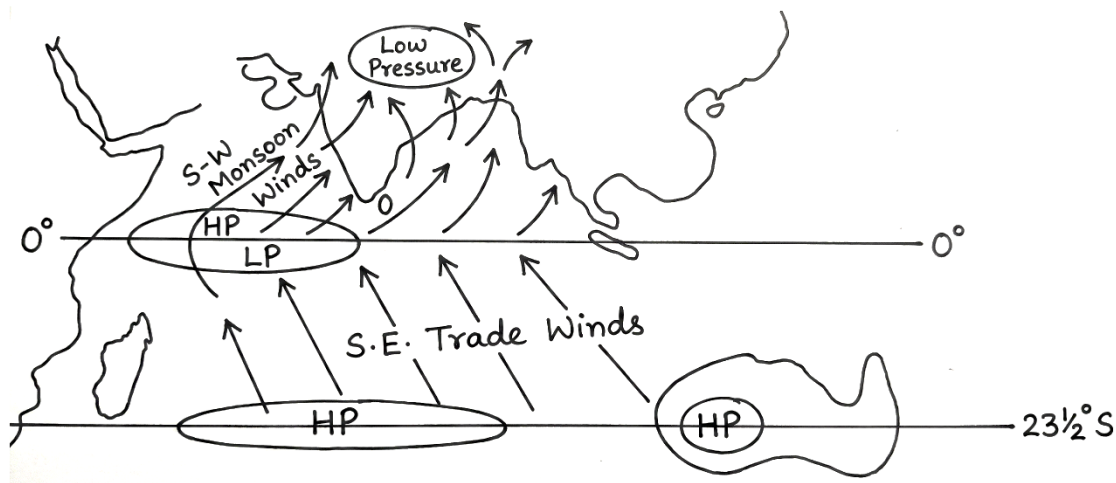
THERMAL EFFECT THEORY

Basis: Hadley's Thermal Cycle. Developed by multiple British climatologists and the Royal Meteorological Institution.

Mechanism:

- When sun's rays are perpendicular over the Northern Hemisphere, intense thermal low pressure forms over the Indian subcontinent.
- This is so intense that equatorial low pressure starts acting as high pressure in comparison.
- Trade winds from the Australian continent cross the equator, turn right following Ferrel's rule, and blow towards India as the southwest monsoon.
- These winds travel ~6,000 km of sea, absorb moisture, and split into two branches over India.
- **Arabian Sea branch:** Heavy rainfall on western slope of Western Ghats.
- **Bay of Bengal branch:** Heavy rainfall in Andaman & Nicobar Islands and north-eastern India.
- As winds move west and north-west, rainfall decreases as moisture diminishes.

Criticism: If origin depended purely on sun's fixed northward/southward movement, monsoon should be predictable. But monsoon is highly uncertain and irregular — so most scientists reject this theory.



Thermal Effect Theory

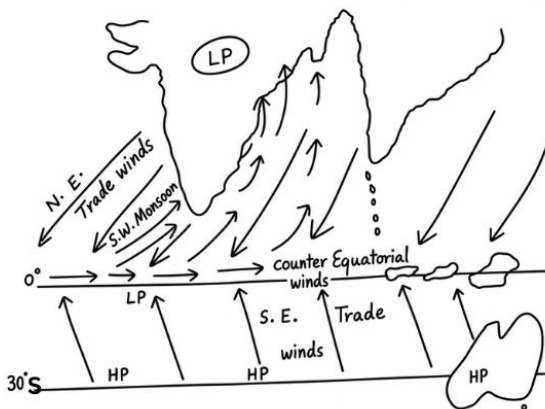
EQUATORIAL WESTERLY WIND THEORY

Proponent: Flohn (meteorologist).

Key Idea: Monsoon wind is actually an equatorial westerly wind arising from inter-tropical convergence (ITC). The westerly wind generated on the Arabian Sea surface does not flow east — it turns towards the Indian subcontinent due to the thermal effect of the subcontinent.

Difference from Theory 1: This theory treats inter-tropical convergence as the basis, whereas Theory 1 considers the SE trade wind naturally turning into SW monsoon. Otherwise both are grounded in thermal effect.

Criticism: Same fundamental problem — cannot explain why the thermal equator is so uncertain (sometimes reaching Shivalik, sometimes staying between 12°–17° N). Hence, limited credibility.



JET STREAM THEORY

Proponents: Yest (original); Dr. P. Koteswaram (India, major contributor).

Jet Stream Definition: Very fast-moving winds in the upper atmosphere (6–12 km altitude) that act as a cover controlling temperature, pressure and atmospheric changes in the lower atmosphere.

Four Types of Jet Streams

Jet Stream	Latitude Range	Notes
Polar Night Jet Stream	Above 60° (both hemispheres)	—
Polar Frontier Jet Stream	30°–65° (both hemispheres)	—
Western Jet Stream	20°–35° (both hemispheres)	Influences India — winter season
Warm / Eastern Jet Stream	8°–35° (both hemispheres)	Influences India — summer season

Jet streams are stronger in the Northern Hemisphere. The last two (Western & Eastern) control India's monsoon.

Winter Season — Western (Subtropical) Jet Stream

During winter, subtropical/western jet winds blow west to east over the Indian subcontinent. These cold winds push surface air → high surface pressure → dry atmosphere.

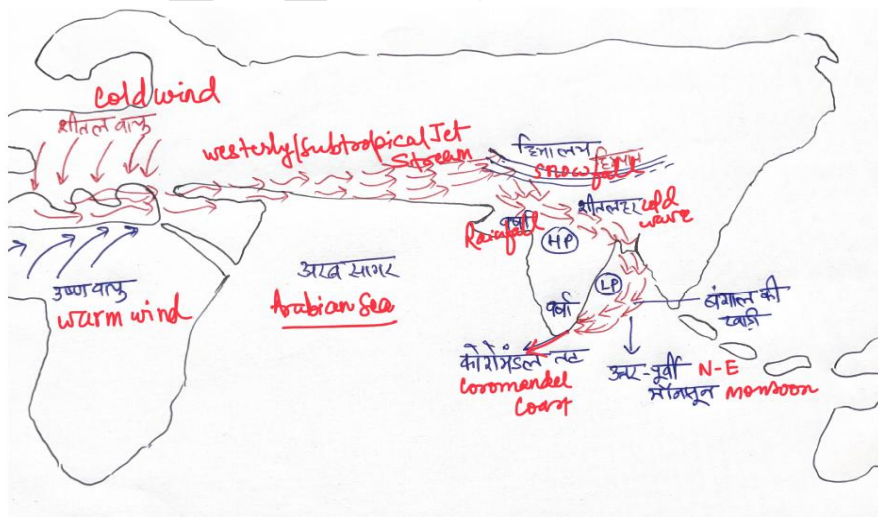
Effects:

- Snowfall in high Himalayan regions.
- **Mahawat (light rain):** In the plains of north-western India (very beneficial for Rabi crops).
- If heavy snowfall in Kashmir → severe cold waves extending to Bihar and West Bengal.
- Comparatively low pressure over Bay of Bengal → winds move continent → sea → dry winter.
- **North-East Monsoon:** Winds reaching the Bay of Bengal follow Ferrel's rule, absorb moisture and cause heavy rainfall on the Coromandel Coast (Tamil Nadu).

India receives only 2–3% of annual rainfall from western disturbances in winter.

Effects of Westerly Disturbances in India

- Act like a temperate cyclone — snowfall in high mountains; rain in NW plains.
- Plains rainfall benefits Rabi crops; mountain snowfall favours apple cultivation.
- Cold waves affect public life in the plains.
- If disturbances occur at Rabi harvest time → significant crop losses.
- Rain usually as drizzle (beneficial); excessive rain can damage Rabi crop.



Winter Season rainfall in India

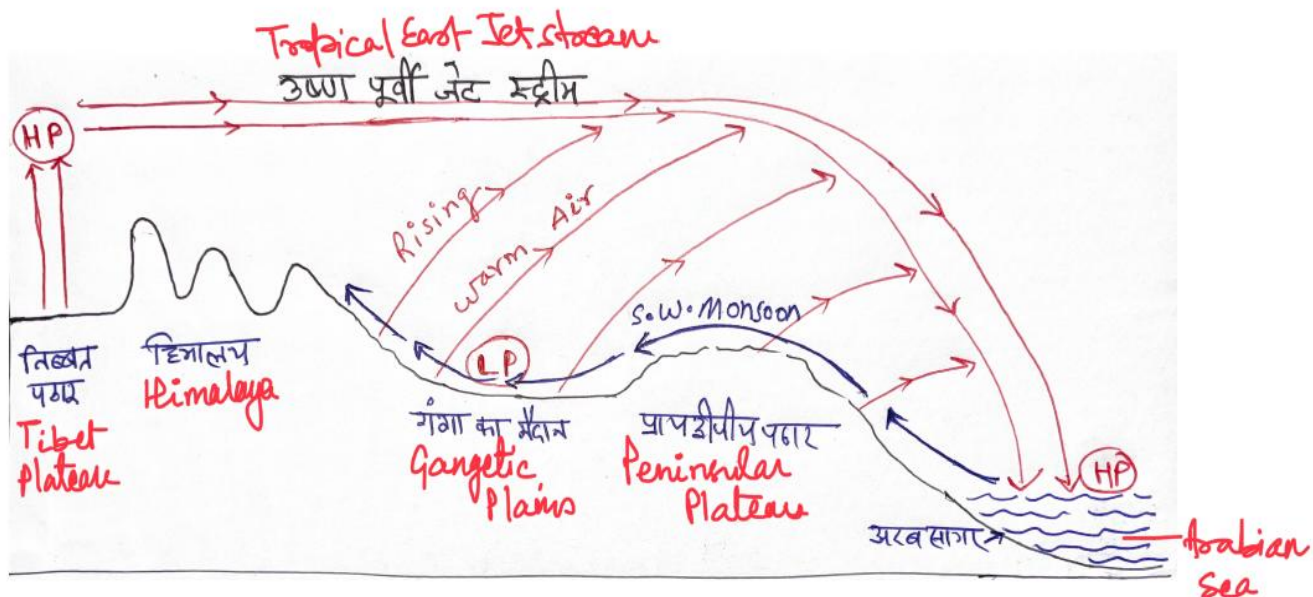
Summer Season — Warm (Eastern) Jet Stream

In summer, warm easterly jet winds blow over the Indian subcontinent while the subtropical jet shifts north.

Origin of Easterly Jet: Solar heat on mountainous and plateau regions of Central and South-East Asia causes hot air to rise vertically; the resulting southward-moving wind follows a south-west direction = easterly jet wind.

Mechanism leading to SW Monsoon:

- Warm easterly jets attract warm surface air → increased thermal effect → subtropical jet displaced northward.
- Large low pressure formed over the Indian subcontinent.
- Easterly jets descend over the Arabian Sea → large high pressure forms over Arabian Sea.
- **Result:** Winds blow from Arabian Sea towards India from the SW direction = **South-West Monsoon**.



Easterly Jet Stream and South-West Monsoon based on the work of Koteschwaram

UNCERTAINTY IN THE SOUTH-WEST MONSOON (MONEX STUDY)

Based on: India-USSR joint research (1973) using the MONEX satellite, which demonstrated sudden heavy rainfall and sudden droughts during the monsoon season.

Key findings:

- **Easterly jet wind is not only seasonal but also irregular** — its intensity varies daily.
- When easterly jets blow over large areas for long periods → low pressure → heavy rainfall → flooding.
- When easterly jets weaken → subtropical jet returns south → high pressure → drought.
- When easterly jets return → new low pressure → monsoon resumes.

Limitation: This evidence-based theory still cannot explain WHY the easterly jet is so erratic. It is attributed to variations in solar heating, but thorough study is still needed.

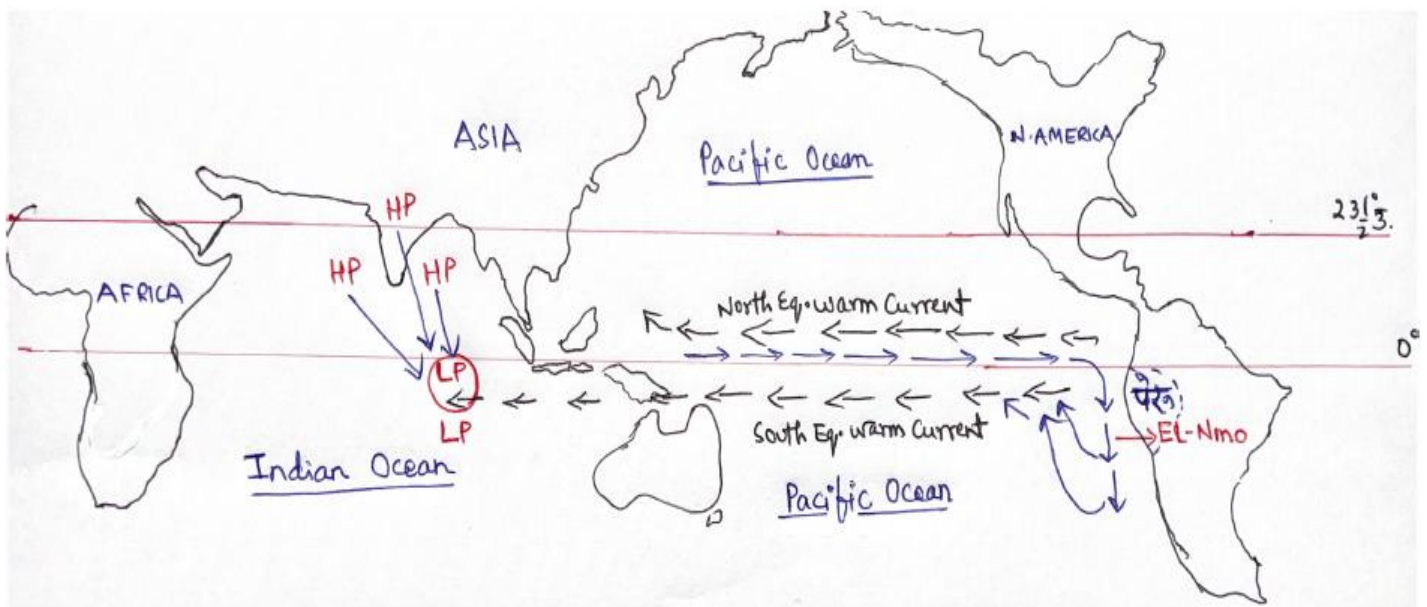
EL NIÑO & LA NIÑA THEORIES

El Niño

Feature	Details
First Studied By	Gilbert Baker (America); idea given in 1924; gained importance after 1987 Indian drought.
What is El Niño?	A sub-surface warm current (also called Counter Current) in the eastern Pacific Ocean.
Location	Flows N→S, 180 km off the Peruvian coast; extension: 3°S to 36°S latitude.
Normal Condition	Cold Peru/Humboldt Current flows S→N near Peruvian coast.
How El Niño forms	In autumn, the equatorial counter current in the Northern Hemisphere shifts south → displaces cold current → warm El Niño current appears.
Effect on Peru	Peruvian coast temperature rises by 8–10°C → many times more rainfall than normal in coastal Peru.

Effect on Indian Monsoon:

- In certain years El Niño extends to the southernmost tip of western South America → warm water pushed east to west.
- Warm equatorial Pacific water moves westward → raises temperature of southern Indian Ocean → creates LOW pressure there.
- In relation to this new low pressure, India, Bay of Bengal and Arabian Sea develop HIGH pressure.
- **Result:** Arabian Sea winds blow towards Indian Ocean low pressure instead of the subcontinent → **massive drought in India.**

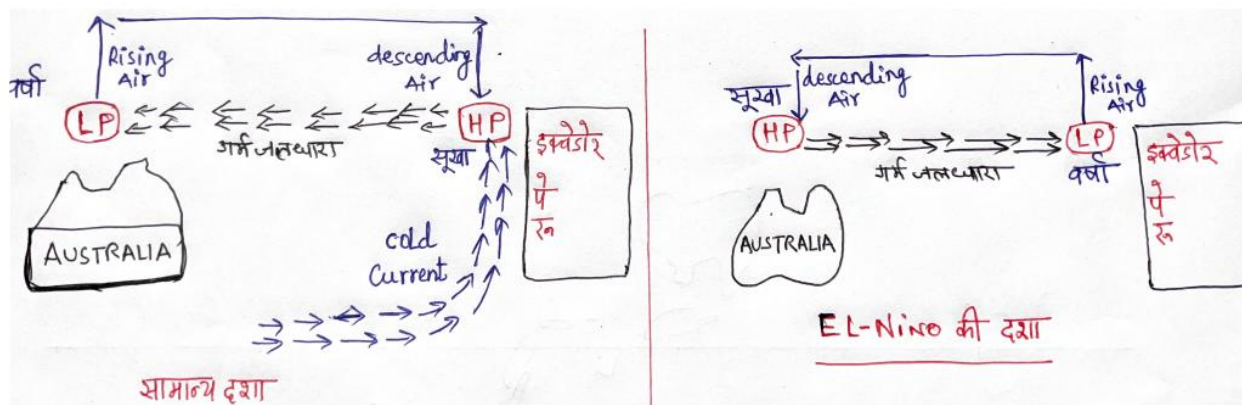


Weakening of the southwest monsoon in India due to El Nino

La Niña

Definition: Called the 'younger sister of El Niño'. Opposite of El Niño. Occurs in the tropical western Pacific Ocean when El Niño's influence in the eastern Pacific ends.

Aspect	El Niño vs La Niña
Trade Winds	El Niño: Weaken → counter-equatorial current towards Peru coast. La Niña: Strong → opposite effect.
Peruvian Coast	El Niño: Temperature rises → heavy Peru rainfall. La Niña: Cold currents rise → high air pressure → dry weather.
Western Pacific	El Niño: Water cools. La Niña: Temperatures rise → increased evaporation → above-normal rainfall in Indonesia.
Indian Monsoon	El Niño: Weakens SW monsoon → drought in India. La Niña: Activates summer monsoon in India → good rainfall.



Southern Oscillation

A meteorological pattern of air pressure changes observed between the Indian Ocean and the Pacific Ocean:

- When Indian Ocean pressure is **HIGH** → Pacific Ocean pressure is **LOW**: SW monsoon in India is more powerful.
- When Pacific Ocean pressure is **HIGH** → Indian Ocean pressure is **LOW**: Greater possibility of weak monsoon in India.

ENSO = El Niño Southern Oscillation — the combined atmospheric and oceanic cycle.

PRE-MONSOON RAINFALL

Rainfall occurring in April–June before the arrival of the southwest monsoon. Regional names:

Local Name	Region / State
Mango Shower (आम वर्षा)	KERALAM and Coastal Karnataka
Cherry Blossom (फूलों की वर्षा)	Karnataka
Coffee Shower	Karnataka
Kaal Baishakhi	Bihar and West Bengal

Norwester	Odisha
Baardoli Chheda	Assam

ARRIVAL & BRANCHES OF THE SOUTH-WEST MONSOON

Arrival timeline: SW monsoon arrives in Kerala by **June 1st** and spreads across the entire country by **July 15th** (~1.5 months to cover all of India). Due to the triangular shape of the peninsula, it divides into two branches.

Arabian Sea Branch — 3 Sub-Branches

Branch 1 — Western Ghats:

- Blocked by Western Ghats → winds climb 900–1,200 m on western slopes → cool down → 250–400 cm rainfall on windward slope and western coastal plains.
- After crossing the Ghats → descend → heat up → humidity lost → nominal rainfall east of the Ghats = Rain Shadow Area.

Branch 2 — Central India:

- Flows north of Mumbai through Narmada and Tapi river valleys → Central India.
- Brings ~15 cm rainfall to Chotanagpur Plateau → enters Gangetic plain → joins Bay of Bengal branch.

Branch 3 — Saurashtra & Kutch:

- Hits Saurashtra peninsula and Kutch → crosses Aravalli and western Rajasthan → very little rainfall.
- In Punjab and Haryana meets the Bay of Bengal branch → combined branches cause rainfall in the western Himalayas.

Bay of Bengal Branch

- Hits Myanmar coast and SE Bangladesh. Arakan Hills deflect a large portion towards India.
- Enters West Bengal and Bangladesh from south and south-east (not south-west due to deflection).
- Branch 1 (westward):** Moves west along the Gangetic plain → meets Arabian Sea branch in Punjab.
- Branch 2 (northward):** Rises north and north-east in Brahmaputra valley → heavy rainfall there.
- Sub-branch (Meghalaya):** Collides with Garo and Khasi Hills → Mawsynram (East Khasi Hills peak) receives world's highest average annual rainfall (>1,180 cm) due to its funnel shape.

Why Tamil Nadu Coast Remains Dry in Rainy Season:

- Tamil Nadu coast lies parallel to the Bay of Bengal monsoon winds (so winds don't strike it).
- It lies in the rain shadow area of the Arabian Sea branch of SW monsoon.

MONSOON BREAK & RETREAT

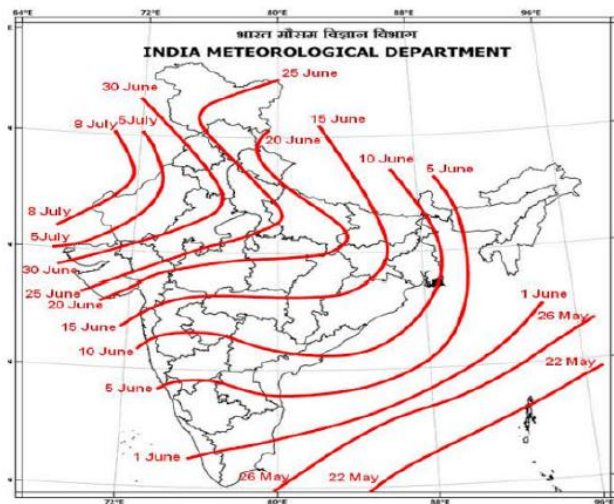
Break in Monsoon

Definition: During the SW monsoon season, after a few days of rainfall, a period of 1–2 weeks (or more) with no rain is called a Monsoon Break. Caused by weakening of the easterly jet stream.

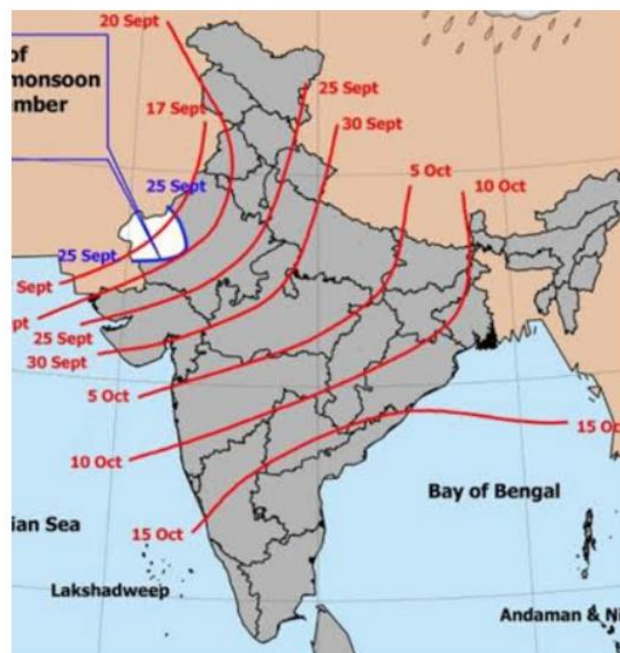
Retreat of Monsoon

October and November = Season of Retreating Monsoon. As the Sun moves southwards after September, the low pressure trough over the Gangetic plains also shifts south → SW monsoon weakens.

Period	Region from Which Monsoon Retreats
First week of September	Western Rajasthan
End of September	Rajasthan, Gujarat, western Gangetic plains, intermediate highlands
Beginning of October	Eastern and north-eastern India
Beginning of November	Karnataka and Tamil Nadu
Mid-December	Low pressure centre completely moves away from the peninsula



Arrival of South-West Monsoon



Retreat of South-West Monsoon

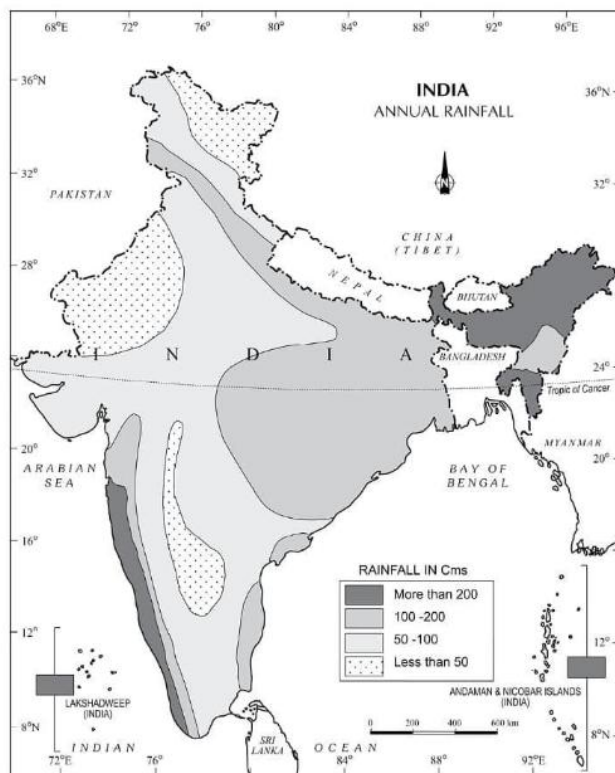
TRADITIONAL SEASONS OF INDIA

Season (Hindi)	Indian Calendar Months	Gregorian Months
बसंत (Spring)	Chaitra–Baisakh	March–April
ग्रीष्म (Summer)	Jyeshtha–Ashadha	May–June
वर्षा (Rainy)	Shravan–Bhadra	July–August
शरद (Autumn)	Ashwin–Kartik	September–October
हेमंत (Pre-Winter)	Margashirsha (Agahan)–Paus	November–December
शिशिर (Winter)	Magha–Phalguna	January–February

DISTRIBUTION OF RAINFALL IN INDIA

Average annual rainfall in India: 115–120 cm (45–47 inches). Varies from <40 cm to >400 cm across different regions.

Category	Annual Rainfall	Regions
High Rainfall	> 200 cm	Western coast, western slope of Western Ghats, sub-Himalayan NE region, Meghalaya hills, Lakshadweep, Andaman & Nicobar Islands
Moderate Rainfall	100–200 cm	Southern Gujarat, eastern Tamil Nadu, Odisha, Bihar, Jharkhand, Chhattisgarh, eastern UP, eastern MP, Cachar Valley (Assam), Manipur
Low Rainfall	50–100 cm	Western UP, Delhi, Punjab, Haryana, J&K, Eastern Rajasthan, Gujarat, Deccan Plateau
Insufficient Rainfall	< 50 cm	Western Rajasthan, Ladakh rain shadow area, parts of South India

**Rainfall Statistics**

Fact	Details
Average annual rainfall	115–120 cm
SW monsoon contribution (June–September)	~75% of total annual rainfall
Pre-monsoon / cyclonic rainfall (April–May)	~10% of total annual rainfall
NE monsoon contribution (October–December)	~13% of total annual rainfall
Westerly disturbances (December–February)	~2% of total annual rainfall
Area with < 75 cm annual rainfall	About one-third of India's total area

Indian Agriculture

Crop Seasons

India has three main cropping seasons: Rabi, Kharif, and Zaid.

Season	Sowing	Harvesting	Key Crops	Region
Rabi (Winter Crops)	Oct–Nov	Mar–Apr	Wheat, Barley, Peas, Gram, Mustard, Linseed etc.	J&K, UP, Punjab, Haryana, HP, Uttarakhand, Rajasthan
Kharif (Monsoon Crops)	June (with monsoon)	Sep–Oct	Rice, Maize, Sorghum, Millet, Pigeon Pea, Cotton, Jute, Groundnut, Soybean	Assam, WB, Odisha, AP, Telangana, TN, Kerala, UP, Bihar
Zaid (Summer Crops)	Mar–Jun	Before Kharif	Watermelon, Muskmelon, Cucumber, Vegetables, Fodder Crops	Pan-India

Food Crops

Food crops are grown on two-thirds of India's total cultivated area. Cereals are sown on about 54% of the total cultivated area. India is 3rd in world cereal production (~11%), after USA and China.

Cereals are classified into:

- Fine grains: Rice, Wheat
- Coarse grains: Maize, Sorghum, Millet, Ragi.

Paddy / Rice

Scientific Name: Oryza Sativa

Season: Kharif crop

Climate: Temperature >25°C; Rainfall >100 cm; Tropical humid

Cultivation Range: Sea level to 2,000 m altitude; 3,000+ varieties

Organic Fertilizers: Blue-green algae, Azospirillum, Phosphobacteria, Azolla

West Bengal grows three varieties:

Variety	Type	Sowing	Harvesting
Aman	Winter Rice (Most Important)	Rainy Season (June–July)	November–December
Aus	Autumn Rice	Pre-monsoon/Summer	September–October
Boro	Spring Rice	November–December	March–April

Key Statistics:

- Largest area under rice cultivation: Uttar Pradesh > West Bengal > Odisha
- Top producing states (2023-24): Telangana > UP > West Bengal > Punjab > Chhattisgarh > Odisha > Bihar
- National average productivity: 2,800 kg/hectare

- Highest productivity in India: Punjab (4,366 kg/ha)
- Highest productivity in World: Japan (7,162 kg/ha)
- Top producing countries: India > China > Bangladesh > Indonesia
- Rice bowl of India: Krishna-Godavari Delta
- Major varieties: Jaya, Jamuna, Kaveri, Padma, Kanchi, Jagannath, Krishna, Vijaya, Annapurna

Wheat

Scientific Name: Triticum Aestivum

Season: Rabi crop

Climate: Temperature 10°C–25°C; Rainfall 50–75 cm; Bright sunshine at harvest

Major Regions: Ganga-Satluj plain (NW); Black soil Deccan region

Key Statistics:

- Largest area under wheat: UP > Madhya Pradesh > Punjab
- Top producing states: UP > MP > Punjab > Haryana > Rajasthan > Bihar > Gujarat
- National average productivity: 3,615 kg/ha
- Highest productivity in India: Punjab (5,069 kg/ha)
- Highest productivity in World: France (7,743 kg/ha)
- Top producing countries: China > India > Russia > USA > France
- Major varieties: Sonalika, Arjun, Kundan, Amar, Bhavani, Chandrika, Kanchan, Gomti

Maize

Scientific Name: Zea Mays

Season: Throughout year (highly adaptable); Maturity 90–150 days

Climate: Rainfall 25–75 cm; Diverse agro-climatic conditions

Special Note: Called 'Queen of Cereals' for highest genetic yield potential

Key Statistics:

- 3rd most important food crop in India (after rice & wheat)
- Largest land area: Karnataka > MP > Maharashtra
- Top producing states: MP > Karnataka > Bihar
- National average productivity: 3,349 kg/ha
- Highest in India: Tamil Nadu (6,820 kg/ha)
- Highest in World: USA (10,510 kg/ha)
- Top producing countries: USA > China > Brazil > India > Argentina

Sorghum / Jowar

4th most important food crop; main crop of semi-arid South and Central India.

- Largest land area: Maharashtra > Karnataka > Rajasthan
- Top producing states: Maharashtra > Karnataka > Rajasthan > UP > AP
- Top countries: USA > Nigeria > India > Mexico
- In South India: both Kharif and Rabi; in North India: only Kharif

Millet / Bajra

Drought tolerant; hot and dry climate of western and north-western India.

- Largest area: Rajasthan > UP > Maharashtra

- Top producing states: Rajasthan > UP > Haryana > Maharashtra > Gujarat
- India is the LARGEST producer globally — contributes ~42% of world millet production

Leading states in total food grain production: UP > MP > Punjab > Rajasthan > Bihar

Pulses

Primary source of protein for the majority of India's population. Cultivated in all three seasons (Rabi, Kharif, Zaid). India contributes ~25% of world pulse production and consumes ~27%. Despite being the largest producer, India imports pulses due to high domestic demand.

Imports from: China, USA, Saudi Arabia, Iraq, Indonesia

- Top producing states: Madhya Pradesh > Maharashtra > Rajasthan > UP > Karnataka

Pigeon Pea / Arhar / Tur

Climate: Temperature 20°C–25°C; Rainfall 40–80 cm

Origin: India

Soil: Alluvial soil (Gangetic plain) to black soil (peninsula)

- Top producing states: Maharashtra > Karnataka > UP (Maharashtra alone produces one-third)

Gram / Chana

Season: Rabi

Region: Central, western and north-western India

- Top producing states: Madhya Pradesh > Maharashtra > Rajasthan

BCW BITS - Mission for Aatmanibharta in pulses (2025-31) initiative, with outlay of ₹11,440 crore to achieve self sufficiency through increased production and area with special focus on **tur, Urad and masoor**.

Oilseeds

Main source of fat in Indian diet. Sown on ~14% of total cropped area. Yellow Revolution is related to oilseed production. India is 4th largest oilseed producer in world (after USA, China, Brazil).

Major oilseed crops: Mustard, Groundnut, Soybean, Sesame, Sunflower, Linseed, Castor, Cotton Seed

Major producing states: Rajasthan, Gujarat, Maharashtra, MP, AP, Telangana, Haryana, UP, WB, Karnataka, TN

- Top producing states: Rajasthan > MP > Gujarat > Maharashtra > Karnataka

Mustard

Season: Rabi (subtropical crop); Sown Oct–Nov; Harvested Feb–Mar

Climate: Temperature 15°C–25°C; Rainfall 60–100 cm; Clear frost-free weather

Most important oilseed crop of India. Rajasthan alone produces almost half of total production.

- Top producing states: Rajasthan > Haryana > MP > UP > West Bengal

Groundnut / Peanut

Season: Kharif (tropical crop)

Climate: Temperature 15°C–25°C; Rainfall 50–80 cm; Large amount of gypsum required
India is 2nd in world groundnut production (after China). Gujarat alone produces almost half.

- Top producing states: Gujarat > Rajasthan > Tamil Nadu > AP > Karnataka

Soybean

Season: Kharif; Sown June; Harvested October; 3–4 irrigations required

Climate: Temp 15°C–25°C; Rainfall 40–60 cm; Acidic loamy soil (pH 6–6.5)

Good source of edible oil and protein. MP alone produces almost half of total production.

- Top producing states: MP > Maharashtra > Rajasthan > Karnataka > Telangana

Sunflower

Season: Rabi (most suitable); Maturity 70–100 days

India imports sunflower from Brazil, Ukraine, Russia. Largest producers globally: Ukraine > Russia > Argentina

- Top producing states: Karnataka > Odisha > Haryana > Maharashtra > Bihar

BCW BITS - NATIONAL MISSION ON EDIBLE OILS- OILSEEDS (NMEO-OILSEEDS) INITIATIVE BY CENTRAL GOVERNMENT (2024-25 TO 2030-31), AIMED AT MAKING INDIA SELF- RELIANT (AATMANIBHAR) IN EDIBLE OIL.

Plantation Crops

Tea

Climate: Temperature 25°C–30°C; Rainfall 150–200 cm; Sloping hills

Origin: China (native)

IN INDIA STARTED AT: Brahmaputra Valley, IN Assam — 1840

Tea Board: Established 1 April 1954; HQ — Kolkata

Rich in caffeine and tannins. India is world's 2nd largest producer and LARGEST consumer.

GI Tagged varieties: Assam Tea, Darjeeling Tea, Kangra Tea, Nilgiri Tea

- Top producing states: Assam > West Bengal > Tamil Nadu > Kerala (Assam = 52%)
- Top producing countries: China > India > Kenya > Sri Lanka > Turkey > Indonesia
- Leading exporter: China > Kenya > Sri Lanka > India

Coffee

Climate: Temp 15°C–18°C; Rainfall 150–250 cm; Sloping hilly terrain, loamy/lava soil

IN INDIA STARTED IN : 18th century, Chikmagalur district, Karnataka

Coffee Board HQ: Bengaluru

Varieties: Arabica (best quality), Robusta, Liberica. India produces only 4% of world production but is in demand for quality.

- Top producing states: Karnataka > Kerala > Tamil Nadu > AP > Odisha
- Top producing countries: Brazil > Vietnam > Colombia > Ethiopia > Indonesia > Uganda > India

Rubber

Climate: Temperature 20°C–34°C monthly avg; Rainfall ~300 cm/yr; Min 100 rainy days

Rubber Board HQ: Kottayam, Kerala

- Top producing states: Kerala > Tripura > Karnataka > Assam > TN > Meghalaya (KERALAM = 74%)

- Top producing countries: Thailand > Indonesia > Vietnam > India > China

Coconut

Coconut Development Board: Cochin, Keralam

Highest productivity in India: Lakshadweep

- Top producing states: Karnataka > Tamil Nadu > Kerala > AP > WB > Odisha
- Top producing countries: Indonesia > Philippines > India

Cashew

Cashew Capital of the World: Kollam, Keralam

- Top producing states: Maharashtra > AP > Odisha > Karnataka > TN > Keralam
- Top producing countries: Ivory Coast > India > Vietnam > Nigeria

Cash Crops

Cotton

Family: Malvaceae

Season: Kharif crop

Climate: Temp 25°C–35°C; Rainfall 75–100 cm; 210 frost-free days; bright sunshine; Black soil

Also Called: White Gold

Most important cash crop. Two main varieties:

- Native (Short-Staple / Old World Cotton)
- American (Long-Staple / New World Cotton) — called 'Narma' in Western India

India is the FIRST country to develop hybrid cotton (gives higher yield).

Three major regions: (1) NW India: Punjab, Haryana, N. Rajasthan | (2) W India: Gujarat, Maharashtra | (3) S India: Telangana, Karnataka, TN

- Top producing states: Gujarat > Maharashtra > Telangana > Rajasthan > Karnataka (Gujarat = 27%)
- Top producing countries: China > India > Brazil > USA
- Highest productivity in India: Tamil Nadu

Sugarcane

Family: Gramineae (Bamboo family)

Season: Tropical & subtropical; 10–18 months to mature

Climate: Temp 20°C–30°C; Rainfall 75–100 cm; well-drained fertile soil

Sowing: Autumn: Sep 15–Oct 31 | Spring: Feb 1–Mar 31

India's 2nd most important cash crop. Uttar Pradesh is the 'Sugar Bowl of India'. India is now world's LARGEST producer and consumer of sugar, and 2nd largest sugar exporter.

Supports: Sugar industry, jaggery, paper, ethanol, power generation, compost manure.

Productivity (T/Ha)	Tamil Nadu	Karnataka	Maharashtra	Punjab	Bihar
Tonnes/ha	104.48	90.82	86.88	82.32	62.93 (12th in country)

- Largest area under cultivation: UP > Maharashtra > Karnataka > Bihar > Gujarat > TN
- Top producing states: UP > Maharashtra > Karnataka > TN > Bihar > Gujarat
- Top producing countries: Brazil > India > China > Pakistan > Thailand

Jute (71ST BPSC MAINS QUESTION)

Also Known As: Golden Fiber

Climate: Warm & humid; Temp 24°C–35°C; Rainfall 160–200 cm; Alluvial soil pH 4.8–5.8

Maturity: 8–10 months

2nd most important fiber crop after cotton. India produces ~56% of world jute; Bangladesh is 2nd (25%).

After Partition, 75% of jute-producing areas went to Bangladesh, but most jute mills stayed in India.

First jute mill in India: Rishra, near Kolkata — 1855, by George Acland.

- Top producing states: West Bengal > Bihar > Assam > Meghalaya > Nagaland > Odisha

Tobacco

Tobacco Board HQ: Guntur, Andhra Pradesh

- Top producing states: Gujarat > AP > Karnataka > UP > Bihar
- Top producing countries: China > India > Brazil > Indonesia > USA

Banana

India is the LARGEST producer of banana globally, producing 26.45% of total. However, India's contribution to global export is only 1%.

First exported to Netherlands (first European country) on 9 November 2023 by APEDA.

- Top producing states: Andhra Pradesh > Maharashtra > Karnataka > Tamil Nadu > UP

Saffron

In India: Cultivated on Karewa land in Kashmir; Pampore region = 'Saffron Bowl'

Season: Cultivated June–Sep; Flowers bloom October

Urdu name: Zafran

India is 2nd largest saffron producer (after Iran). Also cultivated in Spain, Greece, Morocco, Italy, China, Afghanistan.

Govt. also started saffron cultivation in Yangyang, Sikkim (NE India trial).

Types of Agriculture

Primitive Subsistence / Shifting Cultivation

Most ancient form; practiced by tribes in tropical forests. Slash-and-burn method. Trees are cut, burnt, land cultivated for a few years, then abandoned when fertility drops. Low productivity due to lack of modern technology.

State / Country	Local Name
Madhya Pradesh	Bebar / Dahiya
Andhra Pradesh	Podu / Penda
Odisha	Pamadavi / Komaan / Barigam
Western Ghats	Kumari

South-Eastern Rajasthan	Vaalre / Vaaltre
Himalayan Region	Khil
Jharkhand	Kuruva
NE States (Assam, Meghalaya, Mizoram, Nagaland)	Jhoom
Manipur	Paamlu
A&N Islands + Bastar (CG)	Deepa
Mexico & Central America	Milpa
Venezuela	Konuko
Brazil	Roka
Central Africa	Masole
Indonesia	Ladaang
Vietnam	Re

Extensive Subsistence Cultivation

Done in areas with high population pressure on land. Labour intensive with heavy use of biochemical inputs and irrigation. Common in developing monsoon countries (India, Bangladesh, China, Indonesia). Landholdings shrink over generations due to inheritance division.

Commercial Agriculture

Grown primarily for commercial purposes (edible + non-edible). Uses modern inputs — HYV seeds, chemical fertilisers, pesticides. Level of commercialisation varies: e.g., rice is a commercial crop in Punjab/Haryana but a livelihood crop in Odisha.

Plantation Agriculture

Single cash crop cultivated over large area solely for commercial purposes. Link between industry and agriculture. Requires large capital and labour. All produce used as raw material in industry. Examples: Tea, Coffee, Rubber, Sugarcane, Coconut, Banana.

Other Types

Type	Description
Double Cropping	2 or more crops on same land in one agricultural year
Inter Cropping	Two or more crops simultaneously. Ex: Wheat-Mustard, Wheat-Gram, Wheat-Linseed
Mixed Cropping	Crop production + animal husbandry together; mainly in densely populated areas
Dry Land Farming	Agriculture without irrigation in areas with <75 cm rainfall (ICAR definition)
Dairy Farming	Milk and milk products produced commercially; milk-producing animals reared using modern methods

Contract Farming

Agreement between farmers (producers) and buyers on terms for production and marketing — price, quantity, quality, timing, and sometimes inputs (seeds, fertilisers, advice).

Benefits to Farmers:

- Easy access to inputs, services and credit
- Better prices; protection from market fluctuation
- Opportunities to learn latest farming techniques

Benefits to Buyers:

- Continuous supply of raw materials conforming to quality standards
- Cheaper than farming on company-owned land
- Production more reliable than open market purchases

Potential Harms to Farmers:

- Loss of flexibility to sell to alternative buyers when prices rise
- Risk of indebtedness from buyer loans
- Risk of buyers refusing to purchase at agreed prices
- Only big farmers tend to benefit; small farmers often excluded

Farmers (Empowerment & Protection) Price Assurance and Agreement on Agricultural Services Act, 2020:

- Agreement before harvesting; maximum term: 5 years
- Pricing must be clearly outlined in the agreement
- Disputes resolved within 30 days via Conciliation Board, SDM, Appellate Authority
- Payment within 3 days of harvesting; buyer personally collects produce from field
- Government target: 10,000 FPOs to ensure small farmers benefit

Green Revolution**Background**

British economic policies intensified famines. Bengal Famine of 1943 killed ~4 million people. Post-independence, food crisis worsened due to 1962 Indo-China war, 1965 Indo-Pakistan war, and two consecutive droughts.

Father of Green Revolution (World): Dr. Norman Borlaug (USA) — developed hybrid fungus-resistant, high-yield wheat in Mexico. Awarded Nobel Peace Prize, US Presidential Medal of Freedom, Congressional Gold Medal; received Padma Vibhushan in 2006.

Father of Green Revolution (India): Dr. M.S. Swaminathan — mobilised farmers to adopt HYV seeds (Mexican wheat) and convinced the Indian government.

Key Politician: Chidambaram Subramaniam, Agriculture Minister under PM Lal Bahadur Shastri

Initial Experiment: Mexican wheat first cultivated on 1 hectare; India then imported 18,000 tons of HYV wheat seed from Mexico

Definition

The rapid increase in agricultural production in the mid-1960s through the use of high-yielding seeds (HYV), chemical fertilizers, pesticides, canal irrigation, and new technology.

Phases of Green Revolution

Phase I (1965–66 to 1980):

- Focus: Wheat (Mexican) first, then rice (Philippine); Region: Punjab, Haryana, Western UP
- Started with IADP and IAAP programmes; main initiative was HYV programme (Annual Plan 1965-66)
- Food production: 25 MT (1950-51) → 33 MT (1965-66) → 100+ MT (1980) — 3× increase in 10 years
- India became self-reliant; transformed from food importer to exporter

Phase II (1980–1991):

- Focus shifted to rice cultivation in areas with >100 cm rainfall: WB, Assam, Bihar, E.UP, Krishna-Godavari Delta, Cauvery Basin
- Productivity improved in Krishna-Godavari Delta, WB, and Bhojpur (Bihar)
- In other parts of E.UP and Bihar: limited success due to lack of land reforms and traditional farming

Phase III (1991–2003):

- Focus: Pulses, oilseeds, millets, and cotton in dry agricultural regions
- Rainbow Revolution launched — aimed to cover all crops, all regions, all farmers
- Planning Commission divided India into 15 agro-climatic regions for sustainable agriculture

Rainbow Revolution

Announced on 28 July 2000 under the new National Agriculture Policy. Aims for overall development of Indian agriculture.

Revolution	Sector
Green Revolution	Food grain production
Golden Revolution	Fruit and vegetable production
White Revolution	Milk production
Blue Revolution	Fish production
Red Revolution	Meat / Tomato production
Grey Revolution	Fertilizer production
Round Revolution	Potato production
Yellow Revolution	Oilseed production
Pink Revolution	Shrimp / Onion production
Silver Revolution	Egg production
Brown Revolution	Organic farming
White Fibre Revolution	Cotton production
Golden Fibre Revolution	Jute production

Positive Effects of Green Revolution

- India became self-reliant and transformed from food importer to food grain exporter
- Outlook of farmers changed — inspired towards market-oriented, capitalist farming

- Indian agriculture emerged as a profitable business for the first time
- Promoted large-scale agricultural mechanisation (tractors, harvesters, threshers, combines, pumping sets)
- Increased employment in both industrial and rural sectors
- Reduced hunger, improved farmer income and livelihoods

Negative Effects of Green Revolution

Socio-Economic Issues:

- Limited to Punjab, Haryana, W. UP, N. Rajasthan, Krishna-Godavari Delta — promoted inter-regional inequality
- Benefits accrued only to big farmers (10+ hectares); small farmers did not benefit — widened rich-poor gap (inter-personal disparity)
- Productivity of pulses, oilseeds, and coarse grains decreased
- Traditional artisans (carpenters, blacksmiths) lost livelihoods, causing migration from rural areas

Environmental Issues:

- Salinisation: ~50% of agricultural land in Punjab, Haryana, W. UP affected by soluble salts due to excessive irrigation
- Waterlogging: serious problem in canal-irrigated Punjab, Haryana, W. UP, and Indira Gandhi Canal areas in Rajasthan
- Soil erosion: forests cut for farmland expansion; soil loosened by chemical fertilisers
- Pollution: chemical fertilisers and pesticides destroy micro-organisms; pollutes soil, water, and air
- Lowering groundwater table: excessive tube well irrigation in Punjab, Haryana, W. UP raises drought risk
- Deforestation: <6% forest cover in Punjab/Haryana; <8% in UP/Bihar

Health Issues:

- Zinc, copper, lead traces found in milk and vegetables (Indian Council of Medical Research)
- Rising cancer cases in Punjab; burning of paddy straw increases air pollution in Punjab, Haryana, Delhi NCR
- Increase in respiratory, asthma, skin, and eye diseases in Punjab, Haryana, Delhi NCR
- Malaria cases rising in irrigated arid/semi-arid areas (waterlogging creates mosquito breeding grounds)

Special Terms:

Blue Baby Syndrome: Excessive nitrogenous fertilizers → increased nitrates in groundwater → consumed by pregnant women → reduced haemoglobin in blood → born child has fewer red blood cells and appears blue

Eutrophication: Excess fertilizers (like urea) → nitrates reach ponds/lakes → algae growth → water turns green → oxygen depletion → threat to fish → pond/lake destruction

Animal Husbandry

The process of rearing, caring and managing animals to obtain animal products like milk, meat, wool, leather, eggs, etc. and for agricultural work and transportation of goods.

Major Breeds

Category	Major Breeds	Best Breed
Cows	Sahiwal, Gir, Tharparkar, Kankrej, Rathi, Lal Sindhi, Haryanvi, Mewati, Malvi	Sahiwal (best indigenous cow breed)

Category	Major Breeds	Best Breed
Bulls/Ox	Nagauri, Malvi, Hallikar, Kangyam, Killavi, Ponwar, Siri	-
Buffalo	Murrah, Bhadawari, Jafarabadi, Surati, Mehsana, Nagpuri, Pandarpuri, Neeli Ravi	Murrah (best breed)

State Rankings

Category	State Ranking (Highest to Lowest)
Highest number of Cattle	West Bengal > Uttar Pradesh > Madhya Pradesh > Bihar > Maharashtra
Highest number of Buffaloes	Uttar Pradesh > Rajasthan > Gujarat > Madhya Pradesh > Bihar
Top Milk Production	Uttar Pradesh > Rajasthan > Madhya Pradesh > Gujarat > Andhra Pradesh
Top Meat Production	Uttar Pradesh > West Bengal > Maharashtra > Andhra Pradesh > Telangana
Total Livestock	Uttar Pradesh > Rajasthan > Madhya Pradesh > West Bengal > Bihar
Highest Animal Density	Bihar

Sheep

- Best sheep breeds found in: Kashmir, Kullu, Chamba, and Kangra valleys.
- Maximum number of sheep in India: Telangana state.
- Chokla breed (Shekhawati region, Rajasthan — Sikar, Churu, Jhunjhunu): wool quality equivalent to Merino wool.

Top Wool Producing States: Rajasthan > Jammu & Kashmir > Gujarat > Maharashtra

Goat

- Leading states by number of goats: Rajasthan > West Bengal > Uttar Pradesh > Bihar.
- Products obtained: milk, meat, leather, and wool.
- Main source of meat in India.
- Highest milk-producing goat breed in India: Jamunapari.
- Pashmina wool: obtained from Pashmina goat found in Ladakh.
- Angora goat: originated in Turkey; produces high-quality white fiber called Mohair.

Fruit & Vegetable Production

Category	State Ranking
Total Fruit Production	Andhra Pradesh > Maharashtra > Uttar Pradesh > Madhya Pradesh
Total Vegetable Production	West Bengal > Uttar Pradesh > Madhya Pradesh > Bihar > Gujarat

White Revolution

India is an agricultural country where animal husbandry has unprecedented importance. The commercial development of milk products was started to address increasing population, malnutrition, lack of employment, and growing urban demand for milk.

Development Sequence

Year / Event	Details
1965	Establishment of National Dairy Development Board (NDDB)
1970	Launch of Operation Flood
Result	India became the largest milk producer in the world
Father of White Revolution	Prof. Verghese Kurien

Three Phases of White Revolution

Phase	Period	Primary Objective
Phase 1	1970–1981	Improve milk supply in metropolitan cities
Phase 2	1981–1985	Develop milk cooperatives in all districts of India — rural cooperatives, district unions, and state federations (e.g., Bihar State Milk Cooperative Federation, 1983)
Phase 3	1985–2000	Improve quality, diversity, and competitiveness of milk production

Goals of White Revolution

- Create alternative employment sources in rural areas.
- Meet growing urban demand for milk.
- Eliminate middlemen so farmers receive true benefits.
- Arrange milk processing and preservation at block/centre level.
- Develop modern infrastructure for milk production and distribution.
- Develop superior breeds of dairy animals.
- Develop cooperatives and provide better market access.
- Provide required daily protein to people of India.

Key Achievements

Indicator	Before Revolution	After Revolution (2022-23)
Milk Production	~17 million tonnes (1950-51)	230.58 million tonnes
Per Capita Milk Availability	125 grams/day	459 grams/day
Global Share	-	24.64% of global milk production

Regional Disparity in Per Capita Milk Availability

State	Per Capita Milk (grams/day)
Punjab	1283
Bihar	274
Mizoram	58

Challenges of White Revolution

- Uneven regional distribution — northern/northwestern states gained more; eastern/northeastern states lagged.
- Decline of local breeds due to focus on cross-breeding with foreign breeds.
- Inadequate cold chain infrastructure, transportation, and veterinary services.
- Increasing prevalence of adulterated milk — strict enforcement of Indian Food Safety Act needed.

- Lack of hygiene in milk production leading to poor quality and lower prices.

Blue Revolution

Fisheries play a vital role in India's economy — increasing food supply, generating employment, improving nutrition, and earning foreign exchange. It is also known as the Sunrise Sector.

Launch & Key Actions

Detail	Information
Launched in	7th Five Year Plan (1985–90)
Purpose	Increase fish exports, boost domestic consumption, supply nutritious food at low prices
Agency Created	Fish Farmers Development Agency (FFDA)

Steps Taken Under Blue Revolution

- Improved marine agriculture and fish farming in lagoons and shallow coastal waters.
- Mechanisation of fishing operations.
- Developed fish processing centres, cold storage, and transportation facilities.
- Insurance for fishermen: accidental death/disability cover increased from ₹1,00,000 to ₹2,00,000.
- Lighthouse development in deep-sea areas and weather information services for fishermen.
- Eliminated intermediary roles to enhance credibility of Indian fish internationally.
- Technical training and financial assistance for inland fisheries development.
- Integrated fisheries development at ₹3,000 crore over five years.

Objectives of Blue Revolution

- Increase fish production responsibly and sustainably for economic prosperity.
- Modernise fisheries with focus on new technologies.
- Ensure food and nutritional security.
- Generate employment and export earnings.
- Ensure inclusive growth and empower fishermen and aquaculture farmers.

Key Achievements

Indicator	Data
World Rank in Fish Production	2nd (China > India > Indonesia)
Fish Production 1950-51	7.5 lakh tonnes
Fish Production 2023-24	22 million tonnes
Freshwater Fish Share	~65% of total production
Shrimp Production	Largest producer in world
Aquaculture Rank	2nd (after China)
Livelihoods Linked to Fisheries	~1.5 crore people
Fish Feed Availability	9 kg per person

State Rankings in Fish Production

Category	State Ranking
Total Fish Production (fresh + brackish)	Andhra Pradesh > West Bengal > Gujarat > KERALAM > Tamil Nadu
Freshwater Fish Production	Andhra Pradesh > West Bengal > Uttar Pradesh > Bihar (4th)

Note: Pradhan Mantri Matsya Sampada Yojana is the flagship scheme to promote fisheries in India, aimed at increasing fish production, strengthening the value chain, and enhancing fishermen's income.

Aquaculture

The rearing, cultivation, and consumption of aquatic organisms under controlled conditions. Done in both marine (saltwater) and freshwater environments. India ranks 2nd in aquaculture globally, after China.

Types of Aquaculture

Type	Description
Alga-culture	Cultivation of algae species (micro/macroalgae) for commercial uses. Microscopic algae = phytoplankton/microphytes. Macro-algae = seaweed — used in industry but harder to cultivate at scale.
Fish Farming / Pisciculture	Most important part of aquaculture. Fish reared in lakes, ponds, pools. Key species: Salmon, Cod, Carp, Catfish, Tilapia.
Freshwater Prawn Farming	Raising shrimp in freshwater for human consumption. Similar characteristics and challenges to marine shrimp farming.
Shrimp Farming	Marine shrimp farming for human consumption. India is the largest shrimp producer in the world.

Crop Intensity

Introduction

Increase in crop production is possible in two ways:

- Expanding the cropped area
- Increasing the crop intensity

Since most suitable land has already been brought under agriculture, expanding the net sown area has very limited scope. Therefore, increasing crop intensity is now the primary option.

Definition & Formula

Cropping Intensity refers to the number of crops grown on the same farm in a crop year. It expresses the ratio between Gross Cropped Area and Net Sown Area.

$$\text{Crop Intensity} = \frac{\text{Gross Cropped Area}}{\text{Net Area Sown}} \times 100$$

Term	Definition
Gross Cropped Area	Areas of land that are cultivated for more than one crop
Net Area Sown	The entire area on which crops are grown

India's Cropping Intensity

Indicator	Value
Cropping Intensity of India (2023-24)	155.4% – 157%
Increase since Independence	44%
Highest in India	Punjab (201%)
Lowest in India	Mizoram (102%)

Top 5 States (Descending Order)

Rank	State	Cropping Intensity
1	Punjab	201%
2	Haryana	196%
3	Madhya Pradesh	194%
4	West Bengal	193%
5	Uttar Pradesh	182%

State Categories by Cropping Intensity

Category	Range	States
Very High	> 175%	West Bengal, Punjab, Tripura, Madhya Pradesh, Manipur, Haryana, Uttar Pradesh
High	150% – 175%	Himachal Pradesh, Uttarakhand, Gujarat, Rajasthan, Assam, Telangana, J&K
Medium	125% – 150%	Maharashtra (146%), Bihar (148%) , Jharkhand, Karnataka, Tamil Nadu
Low	< 125%	Mizoram (102% — Lowest in India), Goa (114%), Andhra Pradesh (116%), Odisha (120%), Nagaland, Meghalaya, KERALAM

Factors That Increase Cropping Intensity

Irrigation

Irrigation makes it possible to produce and grow crops even during dry seasons, encouraging multi-cropping. It has played a vital role in increasing agricultural production and productivity.

Fertilisers

Fertilisers prevent fields from being left fallow to regain fertility. When fertilisers are used in a coordinated manner, more than one crop can be produced on the same land.

High Yielding Variety (HYV) Seeds

HYV seeds generally take less time to mature, allowing the second crop to be sown earlier. Since the sowing season is limited, these seeds are of great importance.

Mechanisation

Mechanisation reduces the time between harvesting one crop, preparing the field, sowing, and planting. This shorter inter-crop gap makes multi-cropping more feasible.

Crop Rotation

Appropriate crop rotation ensures different crops extract different nutrients from the soil, maintaining fertility. For example, planting pulses (gram, pigeon pea) after rice or wheat returns nitrogen to the soil, encouraging multi-cropping.

Relay Cropping

In relay cropping, a second crop is sown before the first crop is harvested. By the time the first crop is ready to harvest, the second crop is also ripening. This reduces inter-crop idle time.

Indian Example: Gram is sown after sowing the wheat crop. By the time wheat is ready to be harvested, the gram is also fully ripe.

Government Policies Supporting Cropping Intensity

The following government schemes play an important role in increasing cropping intensity:

Scheme	Focus Area
Pradhan Mantri Krishi Sinchai Yojana	Irrigation expansion
Pradhan Mantri Fasal Bima Yojana	Crop insurance
Soil Health Card	Soil fertility management
Traditional Agriculture Development Scheme	Organic/traditional farming
National Mission for Edible Oil	Oilseed production boost
Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)	Direct income support to farmers
Namo Drone Didi	Drone technology in agriculture
Rashtriya Krishi Vikas Yojana	Overall agricultural development
Rainfed Area Development (RAD)	Dryland / rainfed farming support
Per Drop More Crop Scheme	Micro-irrigation / water efficiency
Micro Irrigation Fund	Expanding drip & sprinkler irrigation
National Food Security Mission	Increasing production of rice, wheat, pulses